

frontmatter, the product plan

The final plan before the build. Next.js with Cloudflare R2 and Firestore, every feature free with quantities capped, a configuration panel that sets what each tier allows, thirty-eight screens on desktop and phone, and every number re-derived.

This is the plan the build follows. Revision 6 records two founder decisions taken on 17 September after revision 5 was read: the stack is the Next.js app we already run with Cloudflare R2 and Firestore, not Supabase, and tier contents are set from a configuration panel rather than from constants in the source. Revision 5, which this keeps in full, followed the independent audit of revision 4 (verify/2026-09-17/CLAUDE-AUDIT-REPORT.md, 77 findings) and answers every one of them in verify/2026-09-17/RESPONSE.jsonl. What did not change: the product, the sign-in-first door, the screens' shape, the engine. What changed is in section 0. Every decision still carries its signal: [Z] founders decided, [M] a market or standards page opened and quoted, [R] our own earlier research, [O] measured this session, [L] a constraint, [P] follows from another decision. Sixteen decisions the plan cannot take are in docs/mvp0/DECISIONS-FOR-FOUNDERS-2026-09-17.md, each with a default; this revision is written to those defaults.

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§0 What changed, and why

Revision 6, the founders' two decisions. Both were taken on 17 September, after revision 5 was read end to end.

- **The stack is Next.js, Cloudflare R2 and Firestore** [Z]. Revision 5 recommended Supabase Pro for records and auth, and the founders chose otherwise. It is the cheaper of the two rows section 15 costed, \$43.07 a month against \$65.19 at 1,000 users, and it is what the shipped code already runs, so phase A stops being a migration and becomes a hardening. The structural objections revision 5 raised are carried as build constraints in sections 15 and 18, not as arguments to reopen. Two consequences: the fixed monthly cost falls from \$54.50 to \$29.50, recomputed through section 22, and the proposal to delete `firestore.rules` (F045) is withdrawn, because that file is now live work.
- **Tier contents are set from a configuration panel** [Z], which is the new section 30 and screens S35 to S38. Eleven of the eighteen founder questions stop being decisions before the build and become rows in a table the panel edits. What a panel cannot hold is named there too: a promise made on the sign-in page, an age floor someone has already signed up under, and whether bytes are held at all.

Deferred on purpose, to be settled during the build [Z]: the tagline, the positioning and the product-market read. None of them blocks a phase, none of them changes a screen, and each is better answered with twenty people using the thing than with another research round now.

What revision 5 changed from revision 4, kept here because the audit's corrections still stand.

The three blockers first.

- **Pro's model cost was wrong by a factor of three** (F001). Revision 4 said 100 edits and 5 blueprints on Sonnet 5 cost about ₹120 of a ₹299 month. Re-derived in section 14 with the working shown: ₹383 on Sonnet for everything with the fifteen-file blueprint, against ₹246 net of GST and Razorpay. The default routing is now Haiku 4.5 for edits and Sonnet 5 through the batch API for blueprints, about ₹191, and the price and routing are founder question 2.
- **The free chain carried a link whose data terms were never opened** (F006, F005, F007). OpenRouter's free endpoints are out of the chain until their providers' terms are opened and quoted. Cerebras is named as the 30-day trial it is. The three pools are stated as one shared budget each, and the number of active free users the chain can serve is computed: about 200 after the trial.
- **The legal floor revision 3 carried had been dropped without a word** (F008, F054, F068, F069, F077). It is back as section 23 with an owner and a date per row. The live site sent /privacy, /terms, /pricing and /refunds to the sign-in wall; placeholder pages now serve on the branch and a public-face section names who writes them. The name question is back in section 29.

Then the numbers the fact check marked mismatch or stale, each corrected where it appears: the Doc mode split is 20, 15 and 29 of 64 classified features, with the table in section 7 (F002); Notion's blocks are 27 covered of 32 (F011); the Obsidian request arithmetic is four ship or by design, eleven small, one medium (F012); the research rounds are listed by date and the source count is 244 unique pages (F013); the Drive quota is re-derived with polling included (F027); Craft, Docmost and the cheaper note apps are on the pricing ladder (F004, F059); the slides signal belongs to slides-from-markdown, not to Marp (F061); the Semantic Scholar clause is gone and four API conditions are added (F062); the Model Context Protocol is "a Series of LF Projects, LLC" (F064); the decision cards number 210 (F065); the blueprint is fifteen files and its token cost is re-derived for fifteen (F017, F025); two shortened quotations are quoted in full (F042).

Then the sections the audit said were missing, added: a data model (section 18), a permission matrix (section 19), the format specifications (section 20), performance targets (section 21), the financial model (section 22), the legal floor (section 23), ownership and accounts (section 24), closure of the earlier findings (section 25), the pilot with its stop and

continue lines and the metric definitions (section 28). Phase 0 with the twenty hand-made kits is back (F009). Review state is declared dropped in favour of the change queue (F058). The Yjs exception paragraph is restored beside the no-CRDT rule (F055). Obsidian Multiplayer and Obsidian for Work have risk rows (F060). Max and community are named as Later (F023). Four screens are added: the conflict screen, the AI box with every provider down, the over-cap wall, the Ideas empty state. Every screen change is in section 5 with its finding id.

PART ONE. WHAT WE ARE BUILDING, AND HOW WE GOT HERE

§1 The product in one page

frontmatter is a markdown editor for people whose documents are increasingly written with, and for, AI agents. It is a web app, a desktop app and a phone app on one account. A document is a markdown file that stays the person's own. The editor is the thing we sell. The files never are. That sentence is K1, and the founders confirm or replace it in section 29 [Z].

The promise, in the founders' words. No captchas, no puzzles, no tour. A simple tool with a great editor that feels like home [Z]. The acceptance test for every screen is that a person who knows Google Docs or Obsidian needs nothing explained.

What it does, in six sentences.

1. You sign in with Google or GitHub and land on your documents. Nothing is reachable without an account, exactly as Google Docs works [Z].
2. You write in Markdown mode or Doc mode. Doc mode looks like Google Docs. The file underneath is the same markdown either way.
3. You describe an idea, answer questions at the depth you choose, and get a brief and a blueprint an agent can build from.
4. You share by link, with an expiry or a password, publish a page, or invite people to edit live.
5. You bring in a folder, a Google Doc, a Word file, a Notion export or an Obsidian vault, and push documents to GitHub or keep them in Google Drive.
6. It works offline in the browser, without limits on the desktop, and from the phone's share sheet.

Who it is for. Founders, product people and developers who brief agents. Writers who want Google Docs comfort with markdown files. Obsidian and Notion people who want their notes on the web, shared, and readable by their agents. The audit's reading of the evidence is that only the third group is proven to pay today; section 28's pilot recruits ten of them out of twenty.

What it is not. Not a new format, not a plugin platform in year one, not a chat app with a document attached. Those three were decided and the reasons are in sections 6, 8 and 17. And not the review-state product of the 9 September plan: the change queue on S20 replaces the per-span read state, and section 17 says so.

§2 What we researched, and what it found

The research rounds, by date, all read from the pages named and none from memory [0]: 29 August (the technical PRD, twelve rounds of its own), 6 to 8 September (the pilot reshape and the adversarial round), 9 September (nine lenses), 13 September (the reset, three rounds), 15 and 16 September (twelve agents), 17 September (nine branches). Section 30 lists 244 unique pages.

Why anyone would switch [0]. Obsidian's forum lists 6,051 feature requests. Of its 30 most-liked open requests, frontmatter answers 16: four ship today or are true by design (a web version, a visual editor, one settings set on the account, server-side sync), eleven are small builds and one is medium. Its third most-liked request is a web version, at 949 hearts, and it is absent from Obsidian's published roadmap. Its most-liked request, editing an embedded note in place at 1,078 hearts, is an extension of the Live mode we already ship. Global search and replace, 650 hearts, still does not exist in Obsidian. The plugin registry lists 7,638 plugins with 147,920,815 downloads. Sixty plugins take 59.2 percent of them, and the plan builds those capabilities in. The single fastest-growing plugin is Claudian, an agent panel: 2,112,607 downloads since 5 December 2025. Obsidian's chief executive publishes an agent-skills repository with 48,440 stars since 2 January 2026. And Obsidian's roadmap now lists Multiplayer as Planned and Obsidian for Work as Active [M], which section 27 carries as risks.

What people love and lack [0]. Across the phone-app reviews we counted, clumsy editing was the top complaint at 35 of 156, then sync at 26, slow start at 13 and lost data at 13. Praise went to design at 35, simplicity at 34 and sync at 18. In the lowest-rated Notion reviews, unwanted AI appeared 13 times in 33. The ten switching drivers, by mentions across sources: own files 28, setup burden 24, databases and migration 23, AI edits you cannot switch off 21, price and metering 20, sync 19, speed 19, offline and longevity 19, open source 17, teams 16.

The market split [0]. Everything that stores a person's knowledge for an agent and charges for it keeps the data in its own database, at \$8 to \$375 a month. Everything that keeps knowledge in files the person owns is free and open source. Nobody sells the writing surface.

The standards [M]. AGENTS.md is used by over 60,000 projects. Claude Code does not read it and needs a one-line import in CLAUDE.md. The Model Context Protocol is "a Series of LF Projects, LLC" with a current specification dated 2026-07-28. The skills guide asks for a SKILL.md under 500 lines. These are the formats the blueprint ships in, and none of them is ours to invent.

What the research never did [0]. No customer was interviewed. No prototype was tested with a user. No price was tested. No legal review, no accessibility audit, no security test, no performance measurement of the shipped editor. Section 28's pilot and Phase 0's twenty hand-made kits are the first of those.

This round, nine branches, each from primary pages opened on 17 September [M]:

1. **Free tiers and metering across 40 products.** A cap of 5 free documents sits below every document cap found. The floor is 50. Local-first peers do not count documents at all. Section 13.
2. **The stack.** R2 is right for bytes on every stack. Firestore is the weak piece, for structural reasons rather than cost. Vercel's Hobby plan forbids commercial use, so the first bill is \$20 a seat from user one. Section 15.
3. **Free model providers for the pilot.** Groq, Cloudflare Workers AI and SambaNova's production models state no training. Cerebras is a 30-day trial. Gemini's free tier trains and asks you not to send confidential text. GitHub Models was retired on 30 July 2026. Section 14.
4. **Doc mode against Google Docs.** Of 64 Google Docs features classified, 20 are plain markdown, 15 need an extension, and 29 cannot live in a text file at all. Section 7.
5. **What a markdown document can become.** Slides, mind maps and flow views need no new syntax. Kanban and charts from a table have no open renderer, so those blocks are ours. Section 8.
6. **Public APIs an editor can use.** Twenty were worth keeping, seven of them run entirely in the browser with no key. Section 11.
7. **What people bolt on elsewhere.** Twenty capabilities are top add-ons in three or more ecosystems, and the plan ships every one of them by default. Section 6.
8. **Principles from primary sources.** Nielsen, Apple, Material, PAIR and Martin. Four of five sources argue against sign-in first, one argues for it if Google Docs is the reference class. Section 16.
9. **Sharing, folder upload, Drive and GitHub mechanics.** Password links are a paid feature at Dropbox, Figma and Loom, and Notion has none. Folder upload works in every browser since Safari 11.1 and iOS 18.4. Write-back to a folder works only in Chrome and Edge. Sections 10 and 11.

§3 The decisions, each with its signal

Where it lands | Decision | Signal Front door | Sign in first. No anonymous editing. Google or GitHub, one tap, no password, no puzzle | [Z] the founders chose the Google Docs model. [M] four of five UX sources argue for delaying sign-in, so the sign-in must cost one tap and nothing else Free and Pro | Every editing feature is free. Sharing controls (password links), idea depth (Medium and High), identity (the portfolio) and branding removal are Pro. Quantities are capped on Free | [Z] the founders' rule was "every feature free, quantities capped"; the plan narrows it to these four exceptions and records that in section 29 (F003) Free caps | 50 cloud documents, 1 GB of uploads at 5 MB a file, 5 published pages, 3 live collaborators, 7-day history, 1 GitHub repository with 20 pushes a month, 1 Low blueprint and 10 AI edits a month | [M] 50 is the market floor (Nuclino, Evernote, UpNote); 1 GB is Craft's free storage; 3 is HackMD's "3 invitees" and AFFINE's "Up to 3 members"; 7 days is Notion's, Craft's and AFFINE's history; 20 pushes is HackMD's exact free quota. The founders' candidates of 5, 2 and 1 are below every comparable, section 13 Pro | ₹299 a month or ₹2,499 a year, GST inclusive. Unlimited documents, pages and collaborators, 25 MB a file and 10 GB, 90-day history, password links, 5 blueprints at any depth and 100 edits, the portfolio. Edits on Haiku 4.5, blueprints on Sonnet 5 through the batch API | [Z] price. [M] the first paid tier across peers runs from \$1.99 to \$10; ₹299 is about \$3.12, section 13. [O] the routing is the default that keeps a margin, section 14 and founder question 2 Idea mode | Three depths. Low is free with 10 to 15 questions and a recommendation each. Medium is Pro with 20 to 30 questions, each showing where it stands and what forces the choice. High adds a research pass with sources opened and dated. Not sure is recorded as open, not as a decision | [Z] the founders' Low, Medium and High. [R] the decision-card shape is our own decisions site. [M] PAIR on not laundering the model's choice Doc mode | A Google-Docs-shaped surface inside Live. Ships the 20 lossless features and 15 extensions, refuses the 29 that cannot live in a text file. Font, size, colour, highlight and alignment sit behind More, not in the first level | [M] Google's own rule when exporting to markdown: "Font colors, highlights, and text alignment are removed" (F020) Sharing | Expiring links are free. Password links are Pro. Every published page carries Report, Privacy and Terms | [M] Dropbox, Figma and Loom sell passwords; Bitwarden gives free expiry with a 7-day default. [L] the intermediary duties in section 23 GitHub | Free with a quota, unlimited on Pro, as a GitHub App with the Contents permission, which GitHub grants for the whole repository;

frontmatter writes only under docs/ by its own rule, tested | [M] HackMD's exact model. [M] GitHub: "select the minimum permissions required for the app" (F034) Google Drive | Free. Two-way sync of the files the app created or you picked, with the `drive.file` scope, polled every five minutes | [M] the scope is non-sensitive; at a five-minute poll one project serves 13,201 connected users (F027) Stack | The Next.js app on Vercel, Cloudflare R2 for bytes, Firestore for records and the ledger, Firebase Auth for sign-in, Durable Objects with hibernation for live sessions. Decided by the founders on 17 September, and already what the shipped code runs | [Z] costed in section 15 (F016) AI for the pilot | Free providers whose no-training clause was opened and quoted, in a fallback chain: Groq, Cloudflare Workers AI, Cerebras while its trial lasts, SambaNova. No OpenRouter endpoint. Claude for Pro from day one | [M] Groq, Cloudflare and SambaNova state no training. Gemini's free tier and Mistral's free plan train. OpenRouter's free endpoints carry their providers' terms, unopened (F006) Review | The change queue on S20 replaces the review-state sidebar of the 9 September plan. Attribution stays as a mark in the version record | [Z] the 13 September reset. [R] section 25 closes the earlier claim (F058) Plugins | No in-process plugin system in year one. An API and an MCP server instead, in Later | [M] Obsidian: "cannot reliably restrict plugins to specific permissions". VS Code: "the same permissions as VS Code itself" Representations | Mermaid, Excalidraw, Marp and markmap are embedded. Flow, kanban and table-to-chart are ours as `fm-` blocks. Shapes, never code, from the two copyleft precedents | [M] licences and download counts in section 8 (F063) Portfolio | One markdown file with front matter, served at `frontmatter.in/@handle`. Pro, and late | [Z] late. [M] every site generator wants a folder and a build; a one-file page is the thing nobody offers Phone | Every screen has a phone layout. A bottom bar with five actions, drawers for the tree and the right pane | [M] Material: compact width under 600 dp uses a navigation bar with three to five destinations Legal | The floor in section 23 ships before the first stranger publishes and before the first rupee. Placeholder public pages serve today | [L] (F008) Engine | The twelve invariants of revision 3 stand unchanged | [R] Later, named | A Max tier above Pro; a community as an opt-in index plus GitHub Discussions; the MCP server and API; Team; a custom domain | [Z] asks 27 and 30 (F023)

PART TWO. THE PRODUCT

§4 The journeys

The first five minutes. Sign in with one tap. Land on Home with five ways to start and nothing else. Type in a blank document, or describe an idea, or drop a folder. The document saves as you type. No tour appears. A tip appears only the first time you hover a control you have not used [M] Nielsen: "Tutorials interrupt users, don't necessarily improve task performance, and are quickly forgotten."

Writing. Markdown mode for people who type markdown. Doc mode for people who type in Google Docs. One switch in the Live toolbar. Edit, Live, Reading and Split stay as they are today. The AI box on an empty document, the AI menu on a selection, the problems panel and the instruction-file health panel are the four places the editor helps.

Ideas. A separate tab. Ideas are listed on the left with their state. The idea, its attachments and the industry template sit in the middle. Choose a depth. Answer. The blueprint is written, checked for consistency, and published at an unlisted link with a kickoff prompt that verifies the kit's hash before it unpacks anything.

Sharing. People, a link, a published page. The link can expire, or need a password on Pro. A published page reads without an account, carries a `.md` twin for agents, and carries Report, Privacy and Terms in its footer.

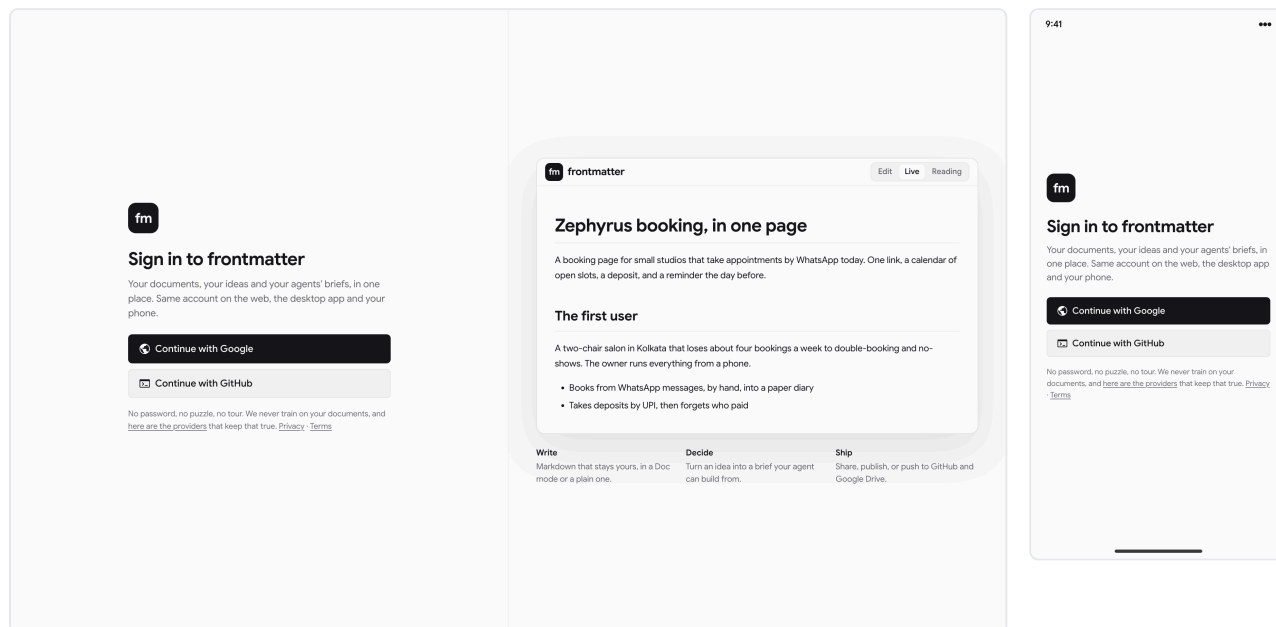
Bringing things in. Drop files or a whole folder. Connect GitHub or Google Drive. Import Google Docs, Word, Notion exports and Obsidian vaults. Nothing is converted unless it has to be.

Everywhere. Offline in the browser with the honest limits stated. The desktop app with files on disk and no document limit. The phone with a share sheet into an inbox note on Android.

§5 The screens

Thirty-eight screens, each on desktop and on the phone. Thirty-four are the product; the last four are the configuration panel of section 30, which only a founder sees. The phone follows the shipped code: a 52 px bar, the editor full width, and the tree and the right pane as drawers. The bottom bar is new. Every screen names what is on it and why, and the finding it answers where it changed in this revision. The screens use two tokens the shipped app does not have yet, an `--ai` blue and the Google Sans Code face; phase B adds both to globals.css (F044).

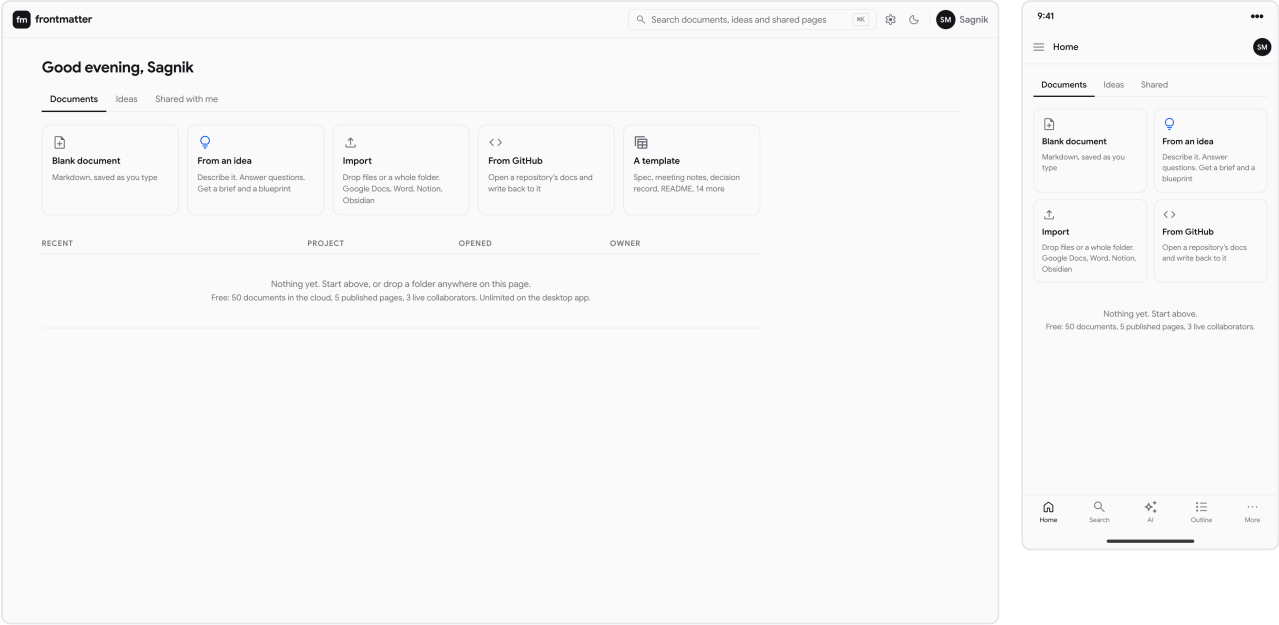
S01. Sign in



- The wordmark, one line of promise, Continue with Google, Continue with GitHub.
- The fine print says what we do not do: no password, no puzzle, no tour, no training on documents, and links the provider list that makes that true.
- Privacy and Terms link to pages that serve without an account (F054). The right half shows the editor once, so the page is not a wall.

Why. [Z] sign-in first. [M] Apple: "People often abandon apps when they're forced to sign in before they can do anything useful", so the sign-in is one tap and the page shows the product behind it. [L] no captcha, ever. The muted text token is retuned to 4.5:1 on every screen (F030).

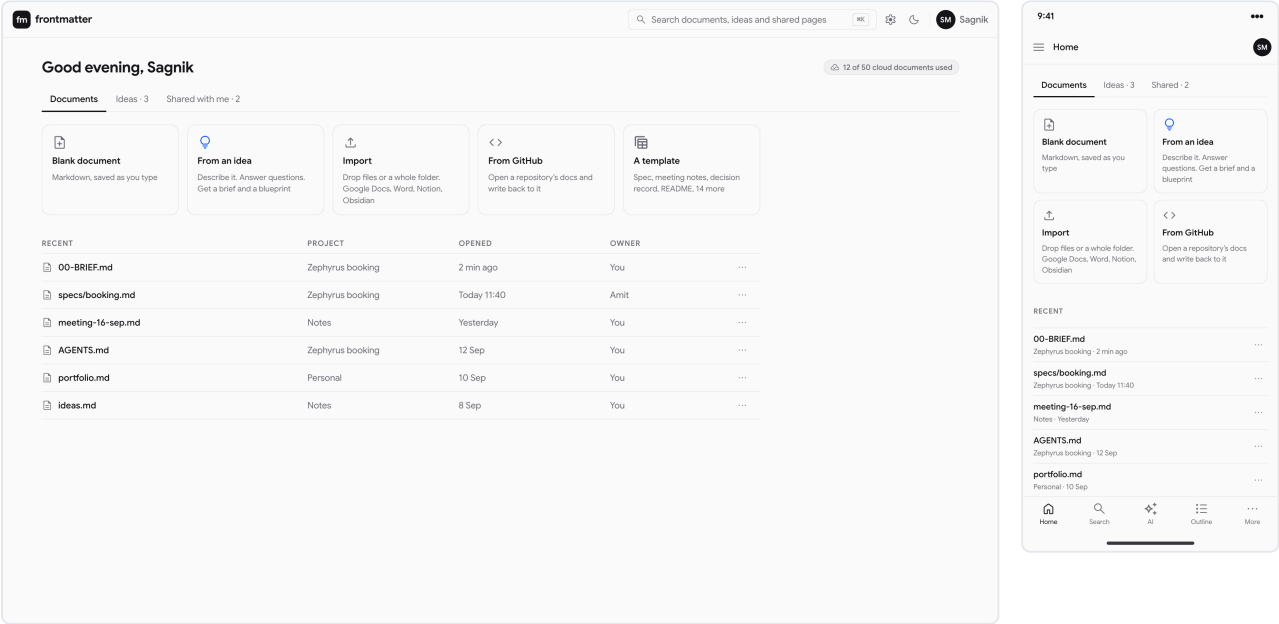
S02. Home, first time



- Five ways to start: a blank document, from an idea, import, from GitHub, a template.
- Three tabs: Documents, Ideas, Shared with me.
- The empty state names the free caps once and says you can drop a folder anywhere.

Why. [M] Nielsen on empty states: "Provide direct pathways (i.e., links) to getting started with key tasks related to populating the empty state." [M] Hick's law: five choices, one of them highlighted.

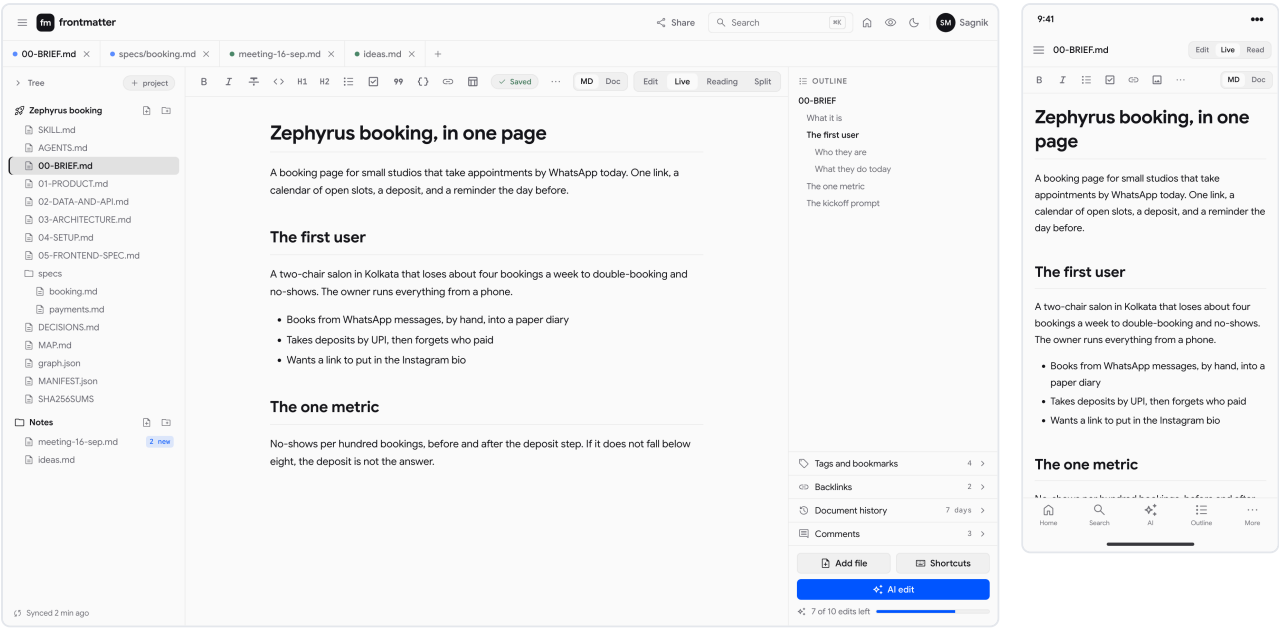
S03. Home



- Recent documents with project, opened and owner. The Ideas tab carries a count.
- A quiet pill shows cloud documents used against the cap.
- The phone shows four starts and the list.

Why. [M] Apple: "Restore the previous state when your app restarts so people can continue where they left off", so the most recent document is first. [P] the caps are visible without a plan page.

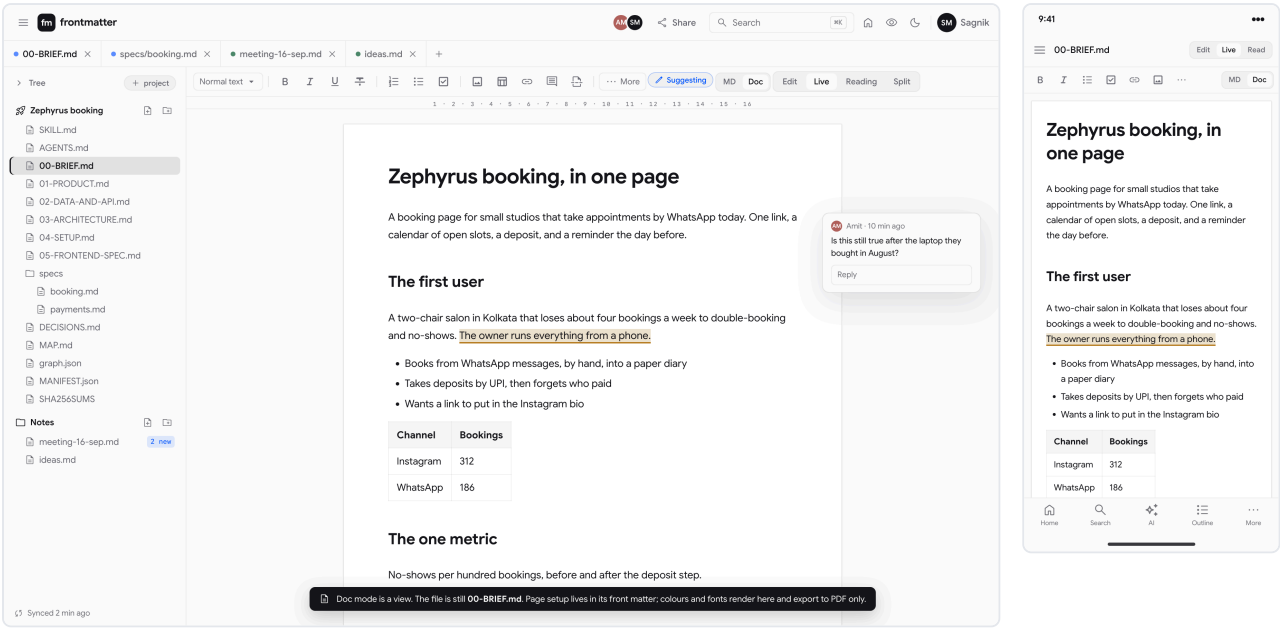
S04. Workspace



- The shipped layout, measured from source: tabs, the twelve-button toolbar, the four modes, the tree and the outline rail.
- The tree shows all fifteen blueprint files (F019). The history row reads "7 days" on Free (F018). The right cluster folds bookmark, search and history into More so the mode segment fits at 1,440 px (F021).
- The phone keeps the toolbar to seven tools and puts the mode segment in the header.

Why. [R] this is md.sgnk.ai as shipped, with the header a Home icon richer. [M] Material: a single pane under 600 dp.

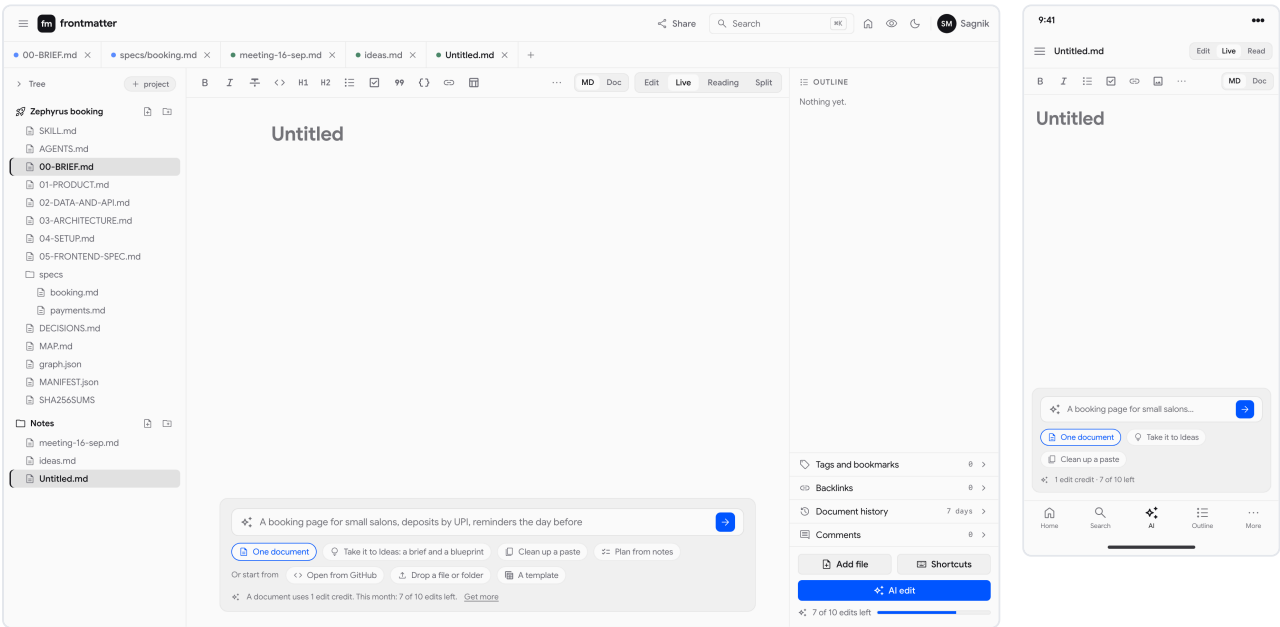
S05. Doc mode



- The Google-Docs-shaped toolbar's first level: style, bold, italic, underline, strikethrough, lists, checklist, image, table, link, comment, page break. Font, size, colour, highlight and alignment sit behind More (F020).
- A paper surface with a ruler. Comments in the margin. Suggesting mode.
- One toast, once: Doc mode is a view; the file is still 00-BRIEF.md; page setup lives in its front matter; colours and fonts render here and export to PDF only.

Why. [Z] "Google Docs features in the doc mode so that people don't have to switch." [M] the honest scope in section 7 says which buttons write markdown, which write front matter, and which do not exist.

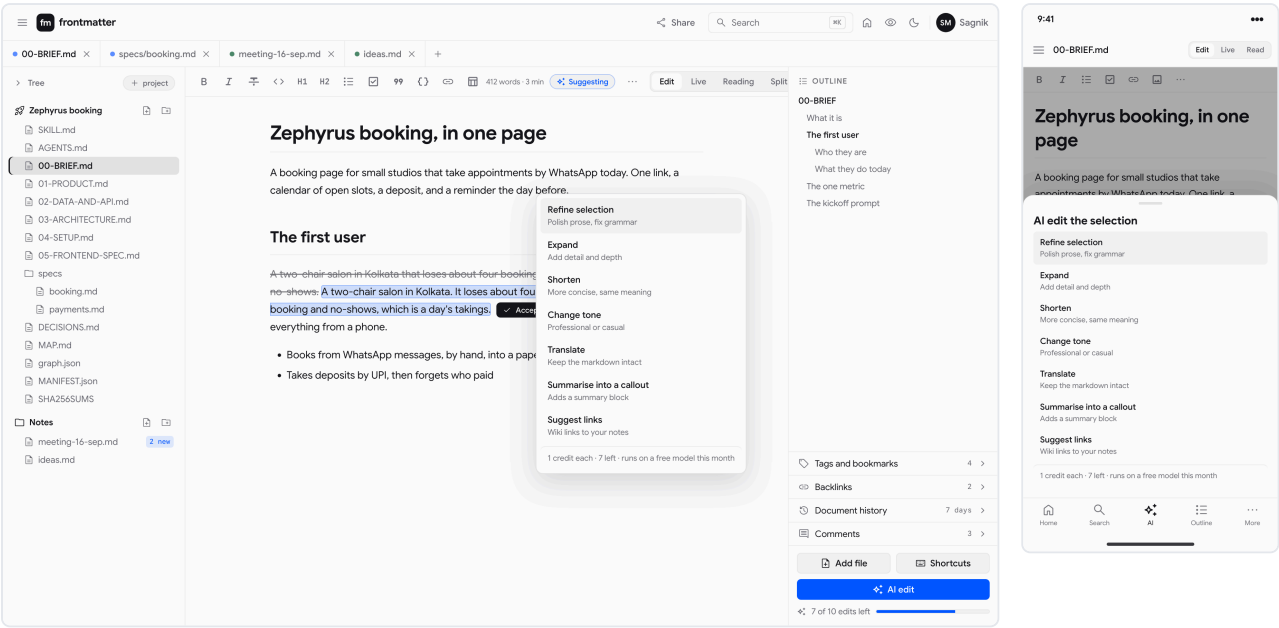
S06. AI writing box



- One box on an empty document. Four chips for the common asks. A second row for the other ways to start.
- The cost is stated before the click: one edit credit, seven of ten left. The rail counts are zero on an empty document (F022).
- Taking it to Ideas is a chip, not a second product.

Why. [R] the funnel is the box. [M] PAIR: "allow users to adapt the output to their needs, edit it, or turn it off."

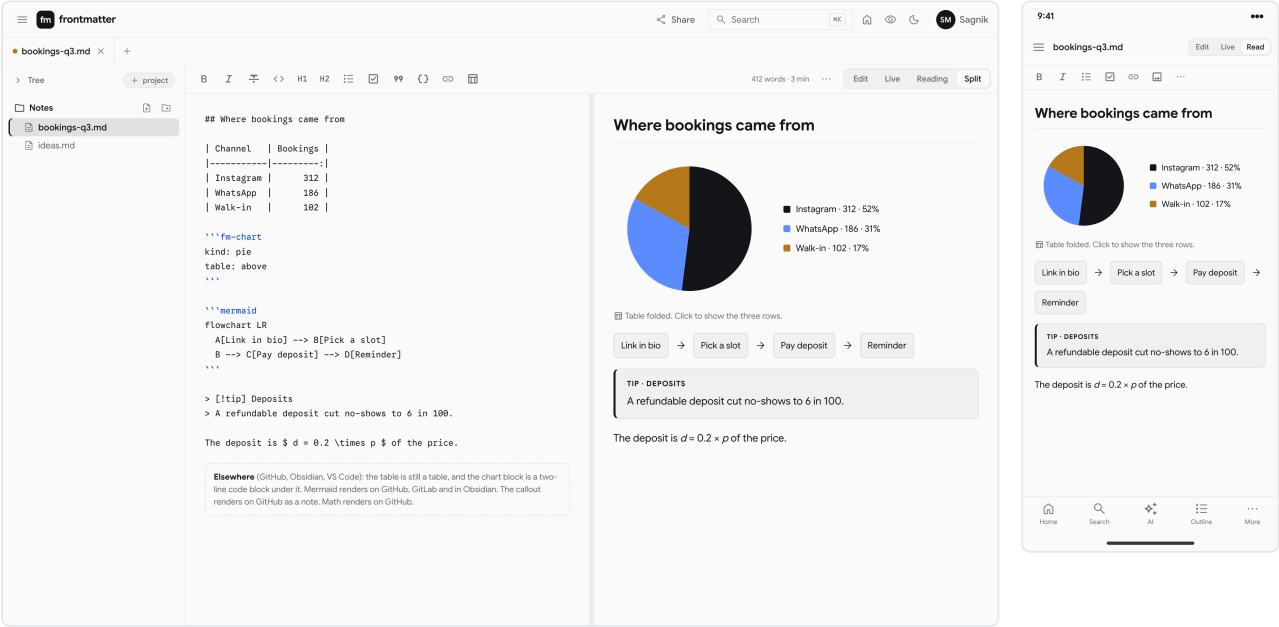
S07. AI edit



- Seven verbs on a selection. The suggestion appears in place with Accept and Reject of equal weight next to it.
- The menu says which provider the month runs on.
- On the phone the menu is a sheet.

Why. [M] Nielsen's direct manipulation: "physical, incremental, and reversible actions whose effects are immediately visible on the screen." [R] the switching driver "AI edits you cannot switch off" at 21 mentions, so nothing is applied without a click.

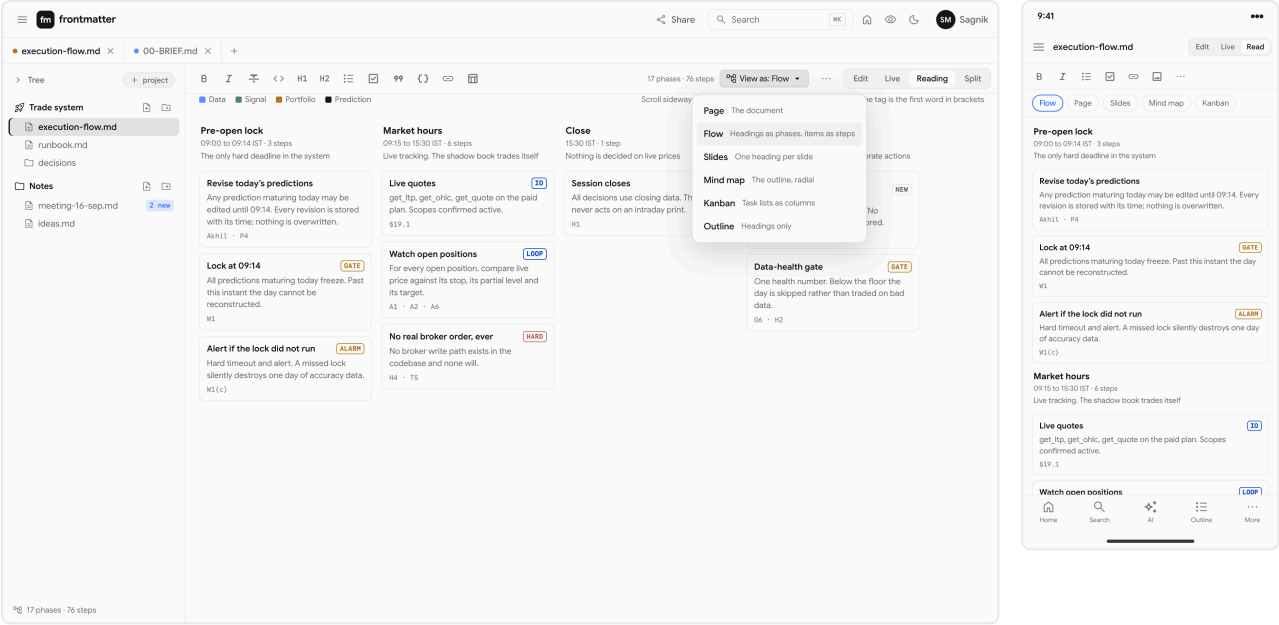
S08. Custom blocks



- Split view. A table, a chart block that reads the table above it, a Mermaid flowchart, a callout, maths.
- The left pane states what each block becomes elsewhere: GitHub, Obsidian, VS Code.
- The phone shows the rendered side.

Why. [R] settled: callouts for prose, fences for data. [M] the table-to-chart pattern exists in the wild in obsidian-charts, whose block points at a block id on the table so the table stays a plain table. The shape is copied, the code is not, because that plugin is AGPL (F063).

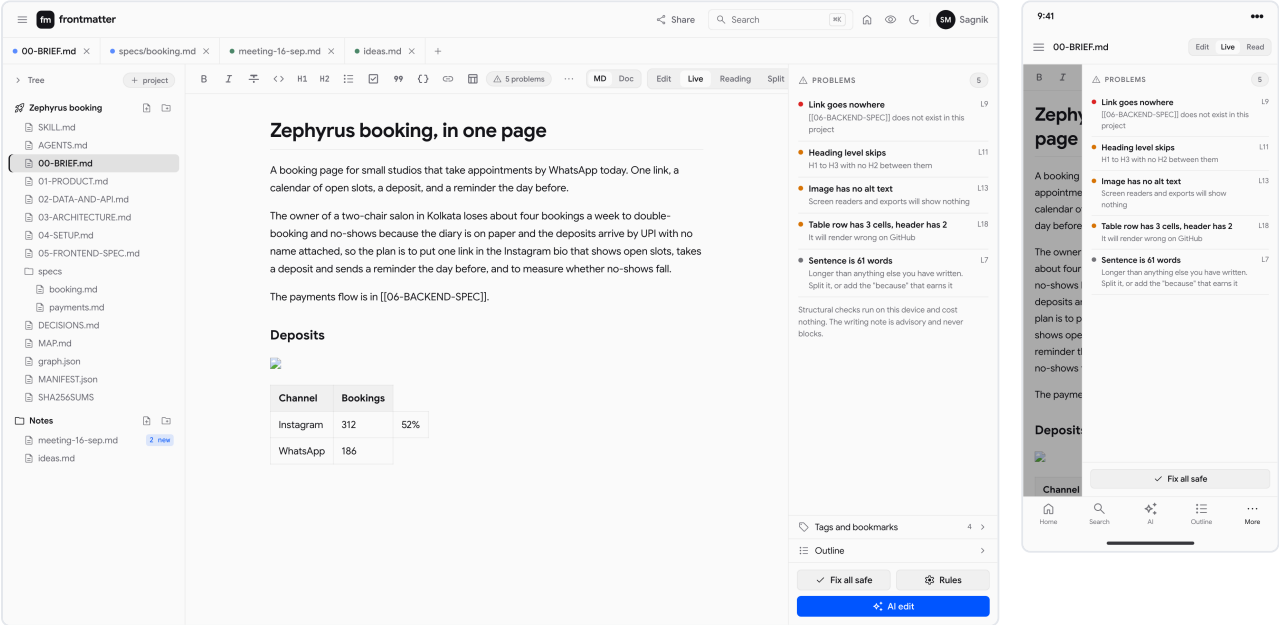
S09. Flow view



- View as: Page, Flow, Slides, Mind map, Kanban, Outline.
- Flow reads the document as phases and steps: an H2 is a phase, an H3 is a step, a bracketed first word is the tag, a trailing line is the reference.
- A legend for the lanes. Scrolls sideways with the arrow keys.

Why. [Z] the founders' flow site is the reference. [M] slides and mind maps need no new syntax: Marp splits on a rule, markmap reads the outline. [P] every view is a projection of the same bytes.

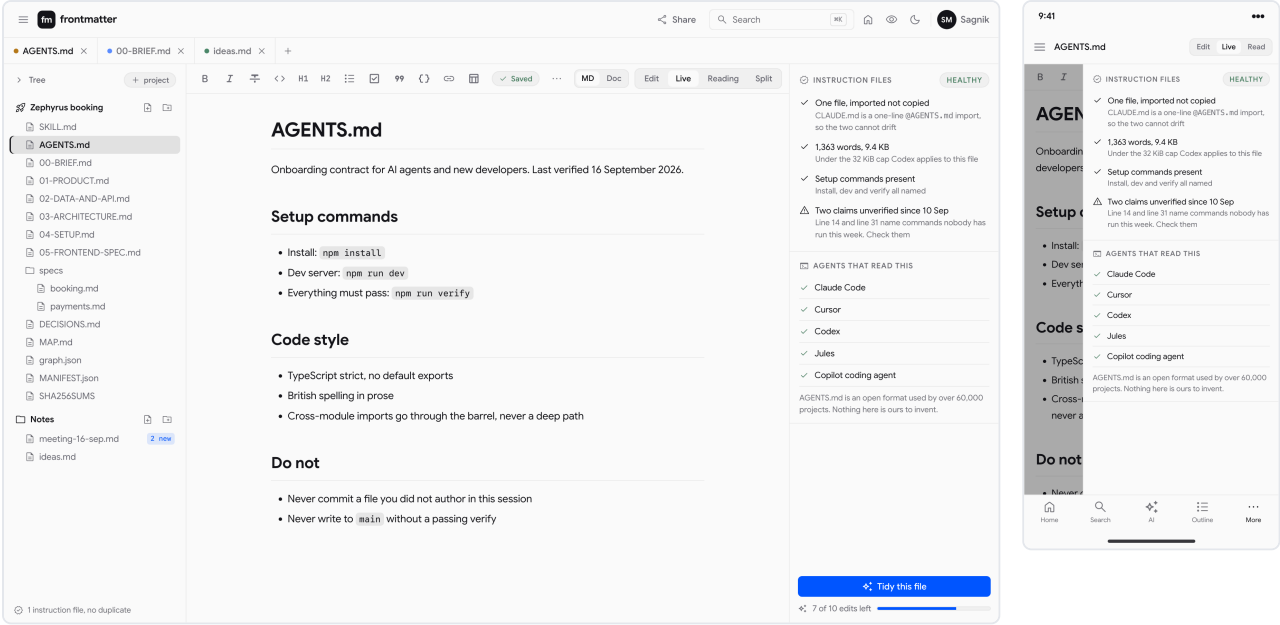
S10. Problems



- Broken links, heading skips, missing alt text, table shape, and one advisory writing note, each true of the document shown (F022).
- Fix all safe. Rules.
- Structural checks run on the device and cost nothing.

Why. [M] markdownlint has 12,180,750 installs on VS Code and Prettier 71,626,148, so a checker and a formatter are expected. [R] the plain-language note never blocks, as our own gate works.

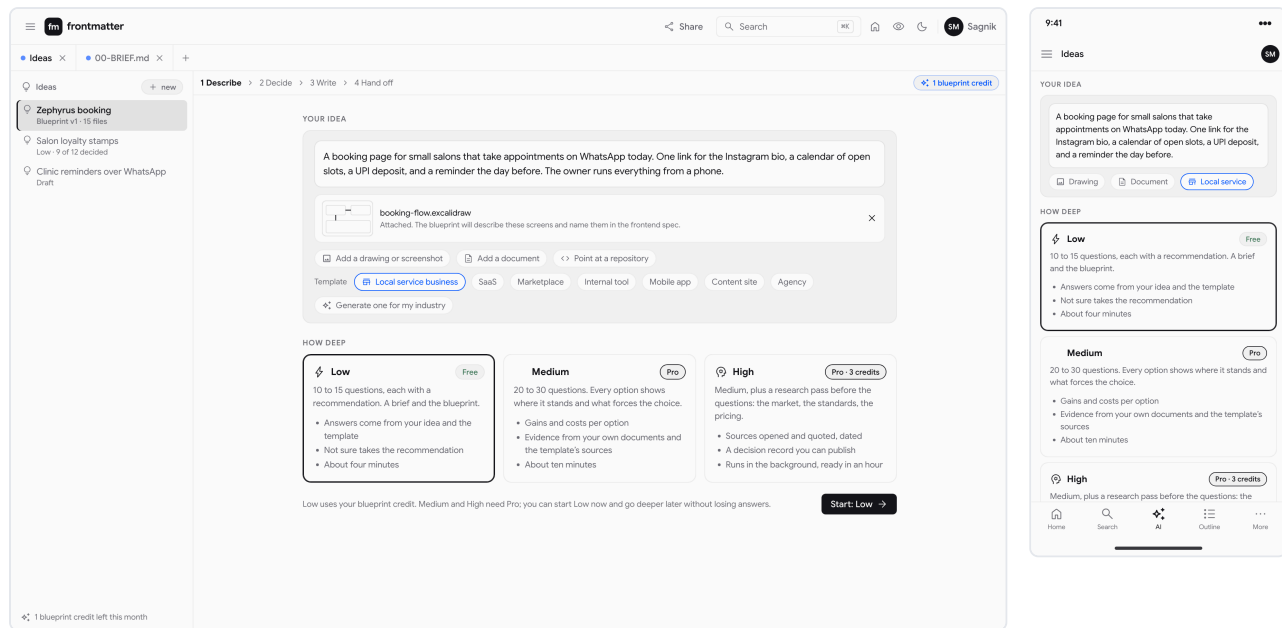
S11. Instruction files



- AGENTS.md with a health panel: one file imported not copied, size under the 32 KiB cap, setup commands present, claims not verified this week.
- The agents that read it. The tree shows the full kit (F019).
- Tidy this file, one credit.

Why. [M] AGENTS.md is used by over 60,000 projects. [0] all six of our own dual-file repositories use the import, so the tool teaches the import rather than diffing two copies.

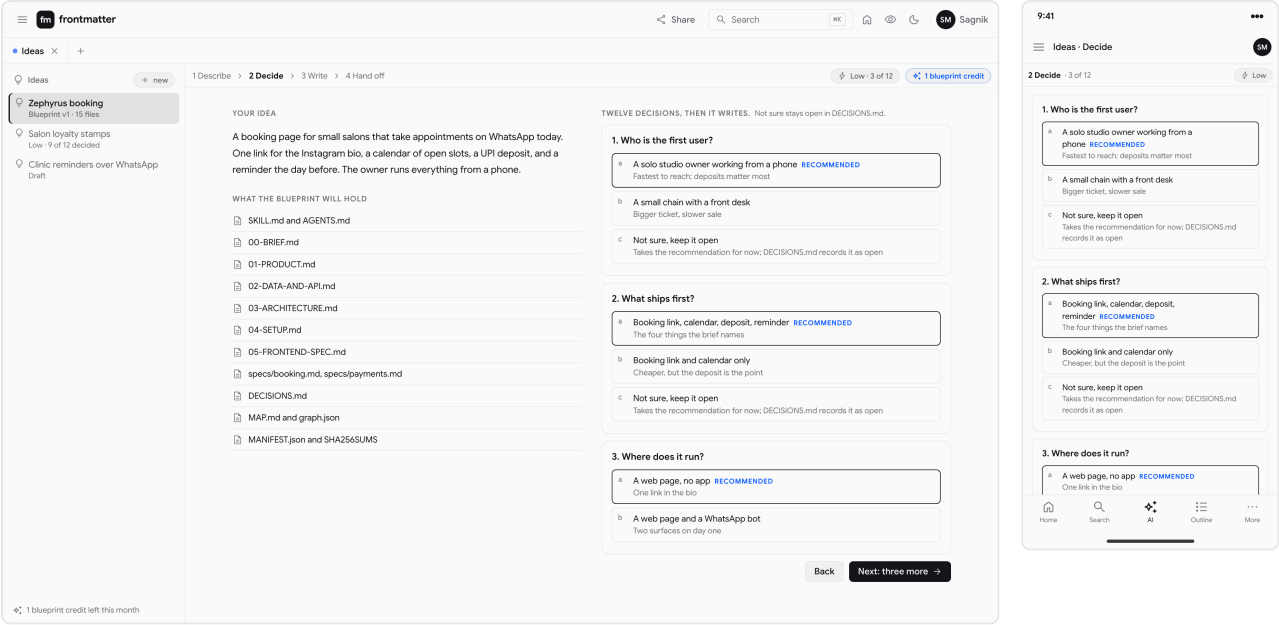
S12. Ideas



- Ideas listed on the left with their state: draft, decided so far, blueprint version.
- The idea, an attached drawing, a document or a repository. An industry template, or one generated for your industry. The drawing is described in the frontend spec the kit now carries (F025).
- The depth chooser: Low free, Medium Pro, High Pro plus credits. Start at Low and go deeper later without losing answers.

Why. [Z] a separate tab with ideas on the left. [Z] the three depths. [M] Nielsen's progressive disclosure has two levels at most, and this is the second.

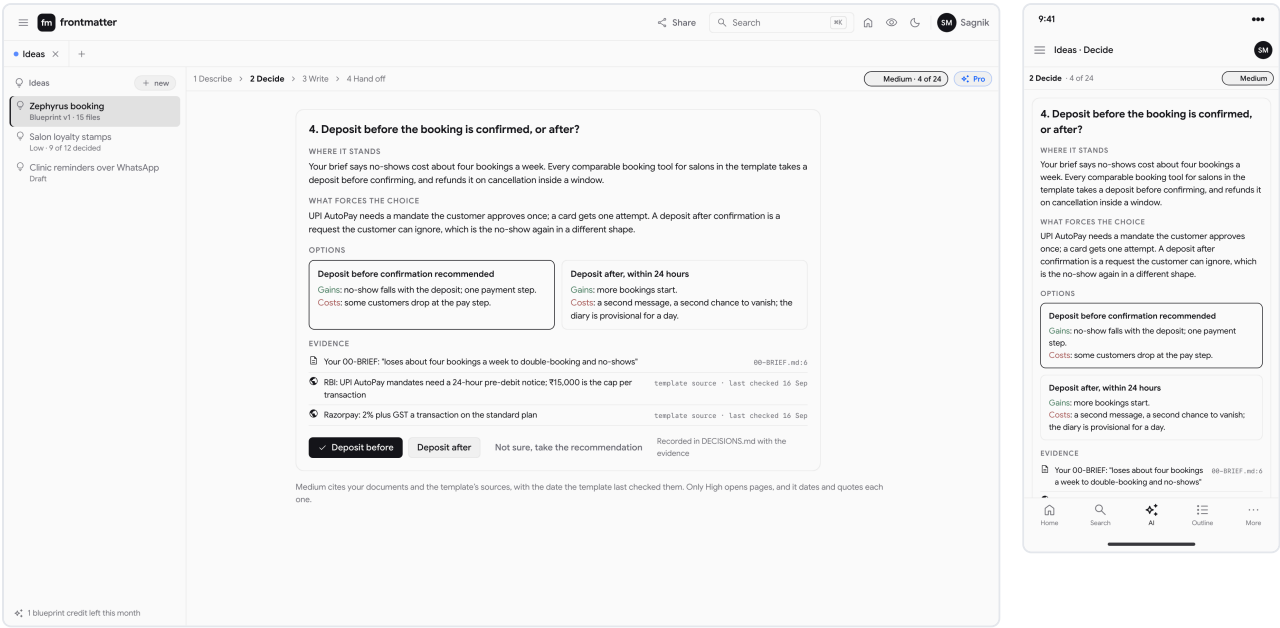
S13. Idea mode, Low



- Twelve decisions in pages of three. Each has a recommendation. Not sure records the question as open and takes the recommendation for now; DECISIONS.md carries it as open, not as decided.
- The blueprint's fifteen files are listed before a credit is spent.
- Steps: Describe, Decide, Write, Hand off.

Why. [Z] 10 to 15 questions on the free plan. [M] PAIR: show alternatives rather than confidence, "Showing multiple options prompts the user to rely on their own judgement."

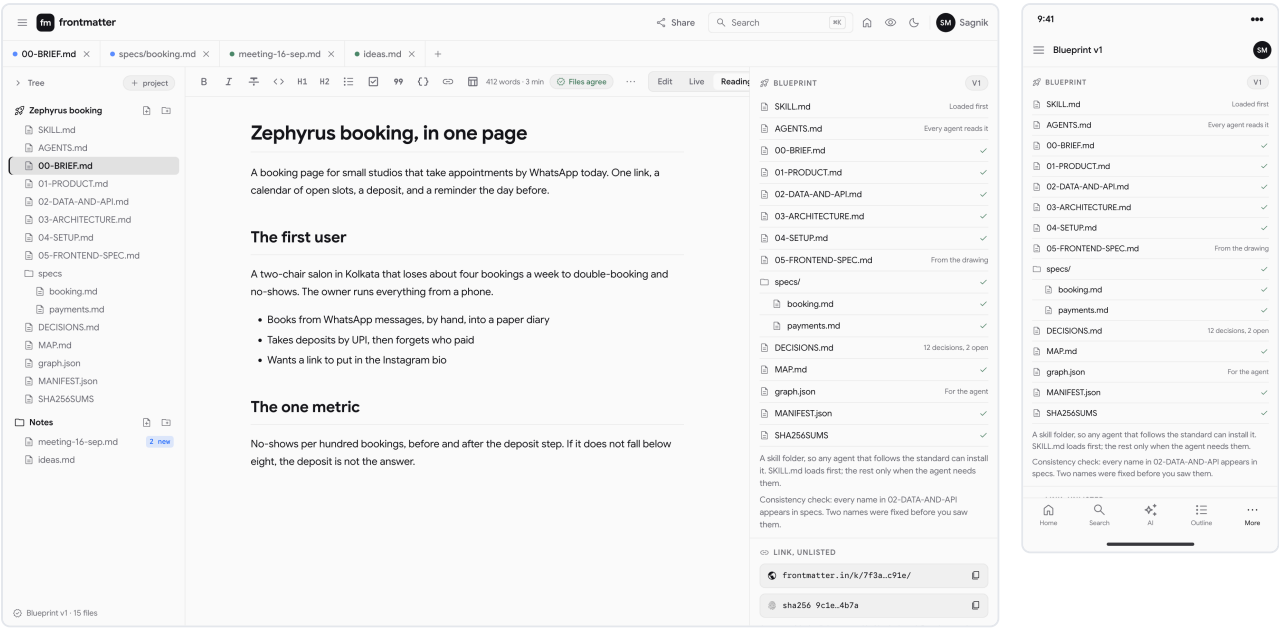
S14. Idea mode, Medium and High



- A decision card: where it stands, what forces the choice, options with gains and costs, evidence rows, the recommendation. On Medium the web rows are labelled as the template's sources with the date they were last checked; only High opens pages (F037).
- The answer is recorded in DECISIONS.md with its evidence.
- High adds a research pass before this step, run in the background.

Why. [Z] "proper insights and all evidence." [R] the card is the shape of our decisions site, which holds 210 decisions.

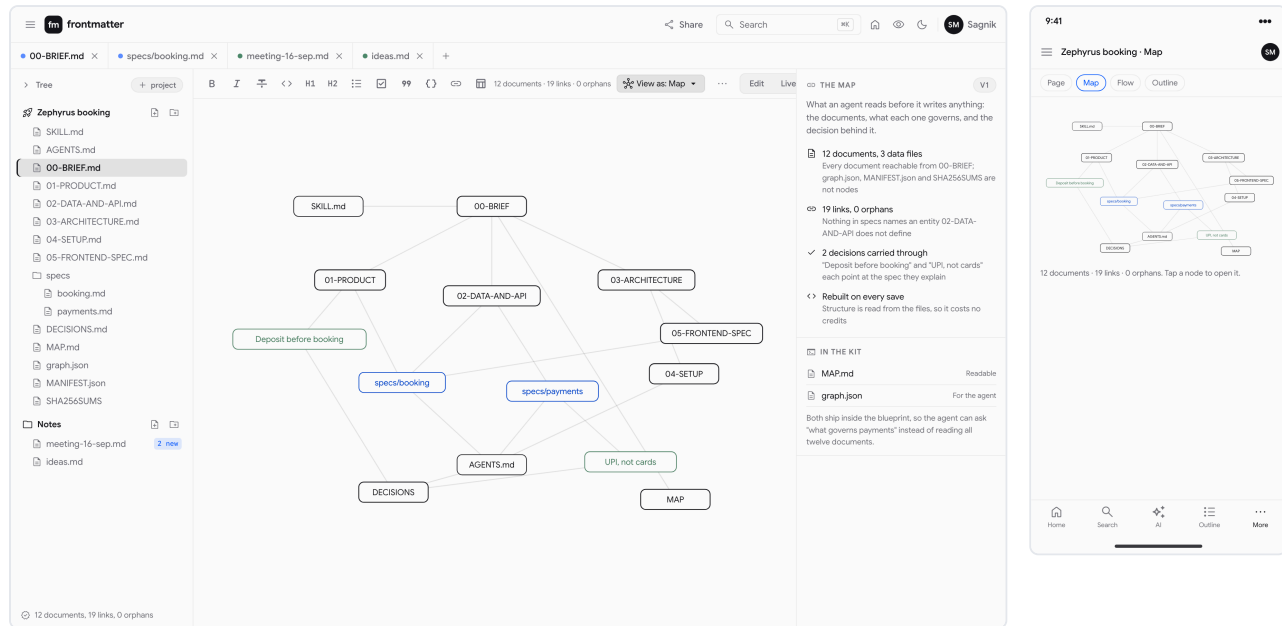
S15. Blueprint ready



- Fifteen files in a skill folder: SKILL.md loads first, AGENTS.md, the numbered documents including the frontend spec, specs, DECISIONS.md, MAP.md, graph.json, the manifest and checksums.
- A consistency check ran before you saw it. An unlisted link. A kickoff prompt for Claude Code, Cursor or Codex that verifies the tarball against the hash printed on this page before it unpacks, and reads before it builds (F028).
- Edit, then publish v2.

Why. [Z] the kit is a skill-shaped folder. [M] SKILL.md and AGENTS.md are the two formats agents read today. [P] a checksum file inside the tarball it verifies proves only that the transfer was intact; the out-of-band hash proves the kit is the one we published.

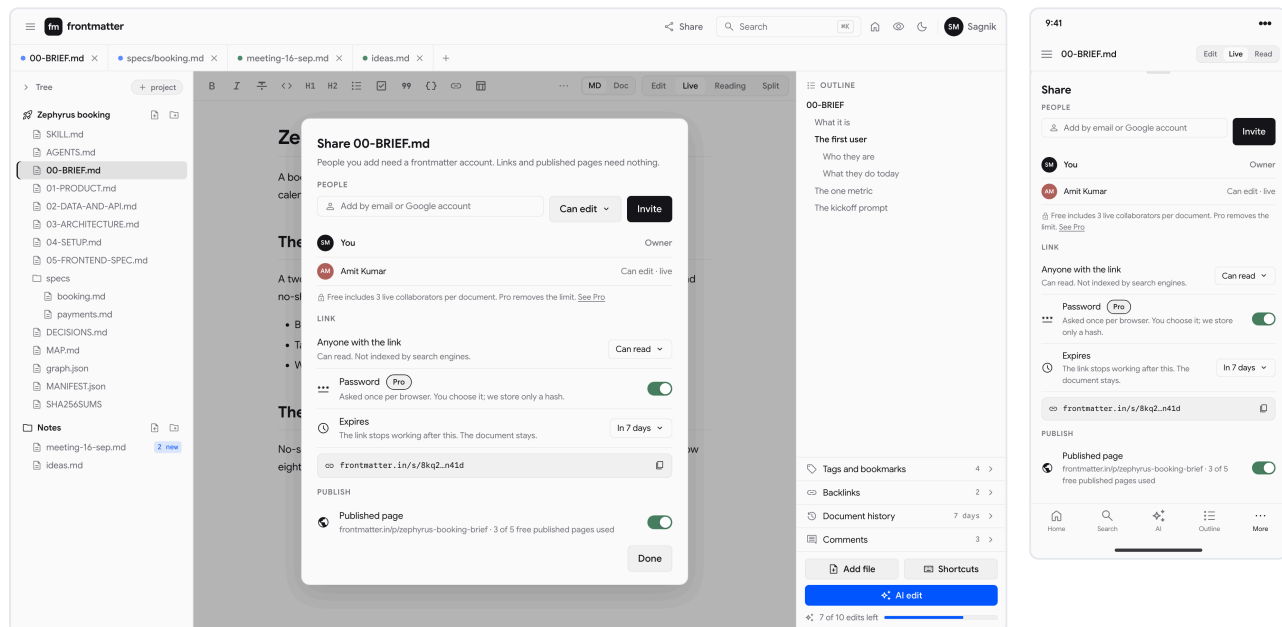
S16. The map



- The documents, what each governs, the decisions behind them, and the instruction file, as a graph. The map counts twelve markdown documents; the three data files are not nodes (F038).
- Rebuilt on every save from the files, so it costs no credits.
- MAP.md and graph.json ship inside the kit.

Why. [R] our own knowledge graph in advox: 4,600 nodes so an agent can query the structure instead of reading 46,000 lines. [P] the map is a projection, never a second source of truth.

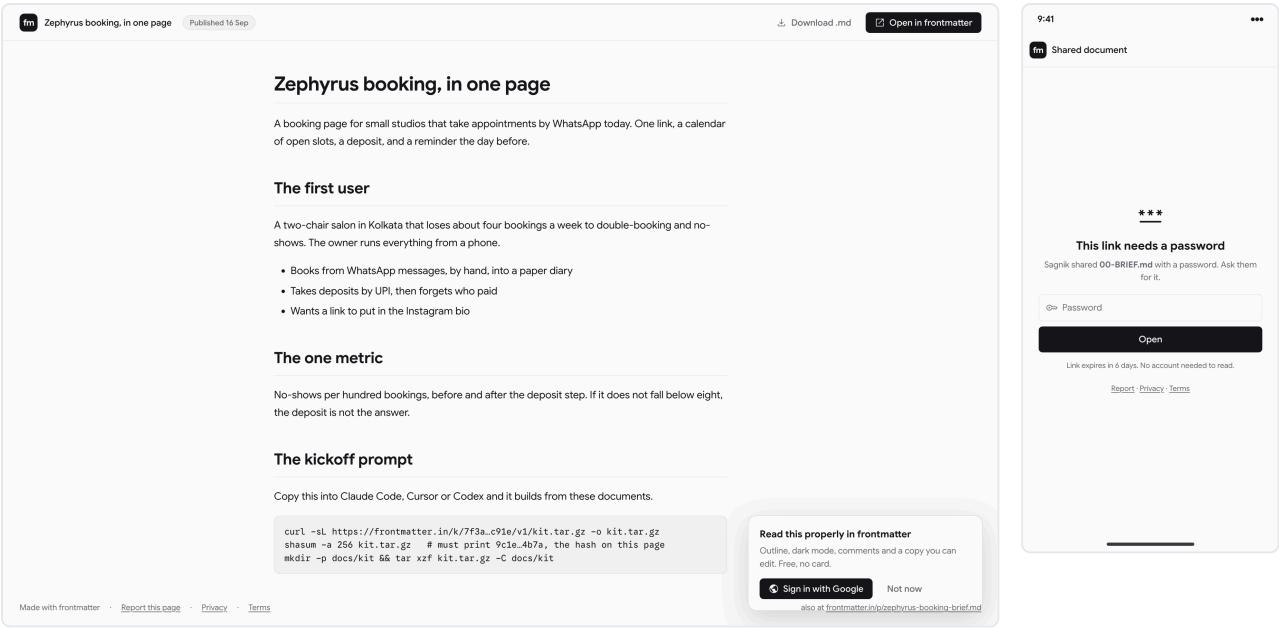
S17. Share



- People, with a role from the matrix in section 19. Three live collaborators on Free.
- A link that can read, expire, or need a password on Pro.
- The published page toggle, with pages used against the cap.

Why. [Z] password-protected sharing. [M] password is paid at Dropbox, Figma and Loom; expiry is free at Bitwarden with a seven-day default.

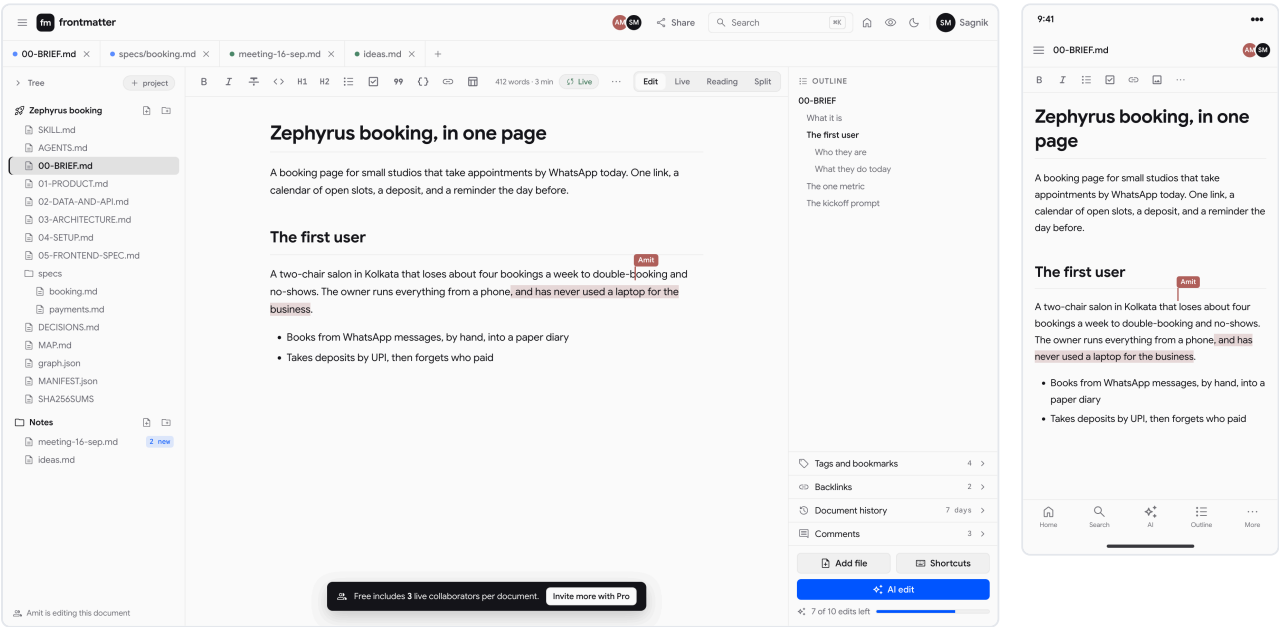
S18. Published page



- Reads without an account. Download the markdown. Open in frontmatter. A quiet card offers sign-in once.
- The footer carries Report, Privacy, Terms and the `.md` twin (F033).
- The phone shows the password gate a reader meets on a protected link, with the same footer.

Why. [R] the published page is the funnel. [M] llms.txt and the `.md` twin pattern are what Anthropic, Cloudflare, Stripe and Vercel serve. [L] a page a stranger publishes needs a report route, section 23.

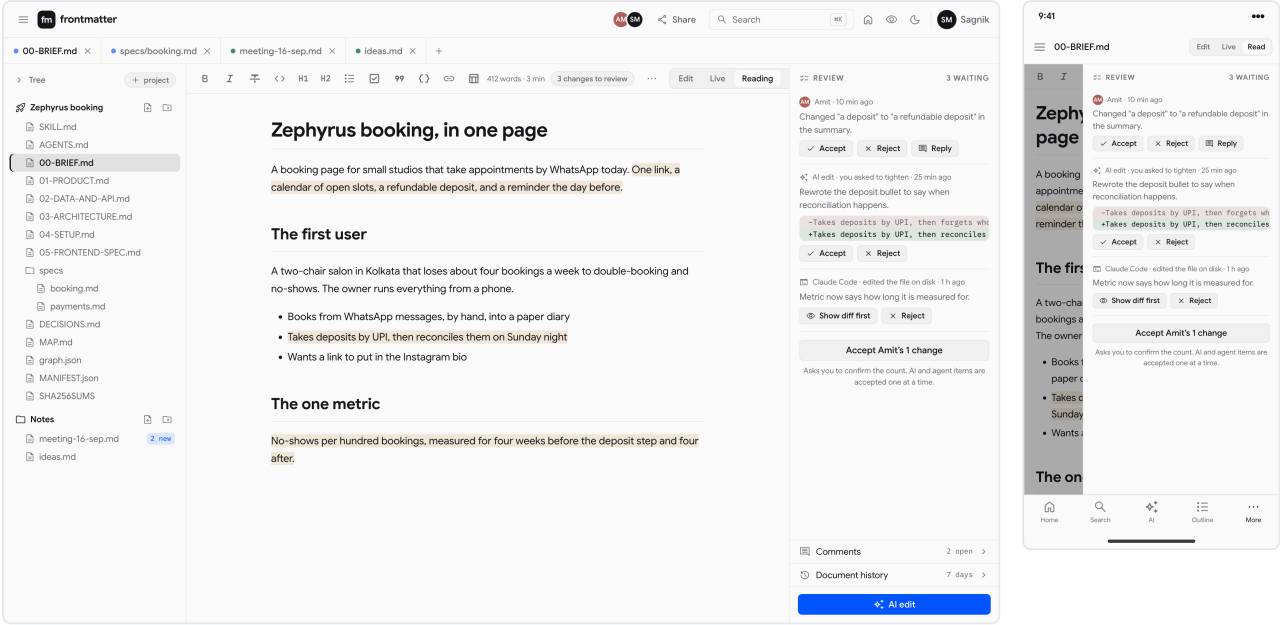
S19. Live collaboration



- Presence avatars, a named cursor, the other person's text highlighted as it lands.
- The toast states the free limit once.
- On the phone, presence sits in the header.

Why. [M] Obsidian's Multiplayer is Planned, not shipped, and section 27 carries it as a risk. [R] the session is the exception the no-CRDT rule allows, section 15.

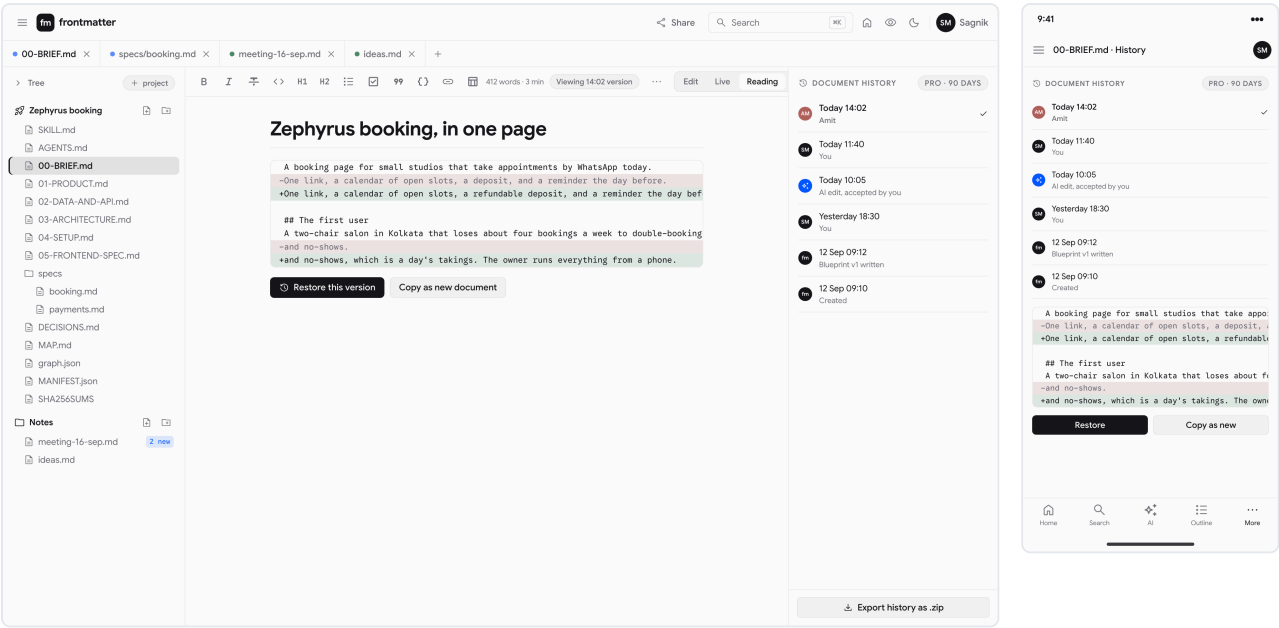
S20. Document review



- Changes waiting, each with who made it: a person, an AI edit you asked for, or an agent that edited the file on disk through the desktop folder.
- Accept, Reject and Reply of equal weight. Accept all applies only to a named person's edits and asks you to confirm the count first; AI and agent items are accepted one by one with the diff shown (F029).
- Changed spans highlighted in the document.

Why. [R] the proposal is a first-class object: agents propose, people accept. [M] Buçinca 2021, Jakesch 2023, Doshi and Hauser 2024 and Draxler 2023 on over-acceptance of AI text, in the note-research report. [M] iA Writer now leads with authorship display, so the market has caught up with the need.

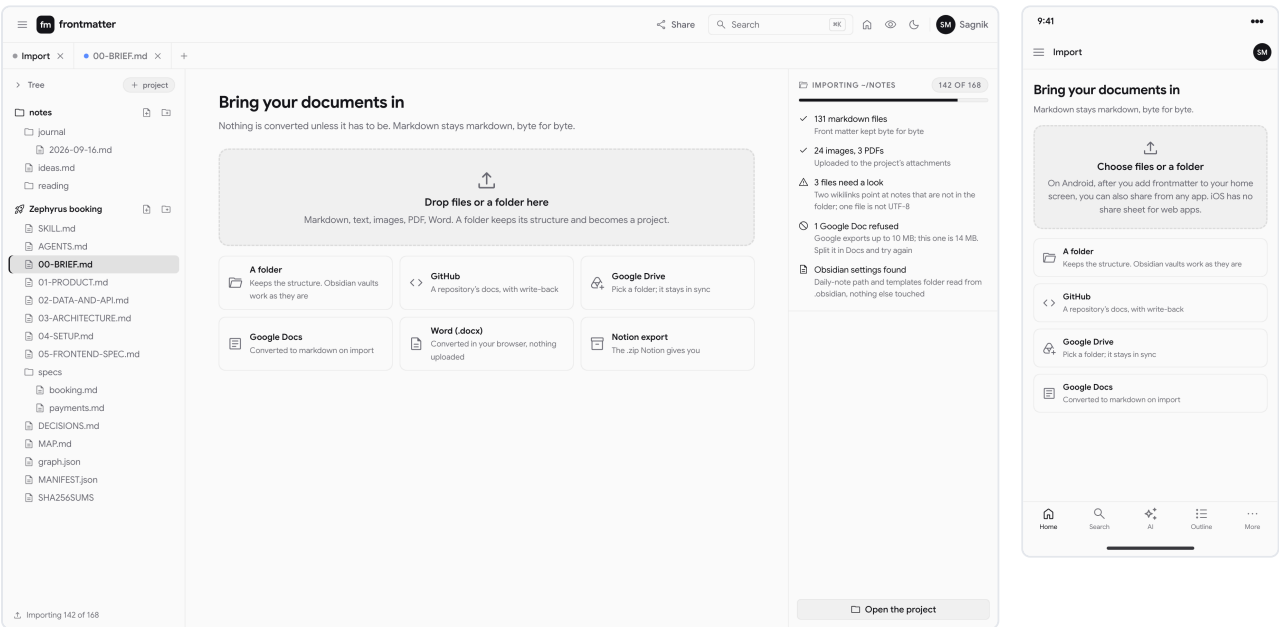
S21. Document history



- Every version with its author, including the AI edit and the blueprint write.
- A diff against the current version. Restore, or copy as a new document.
- Seven days on Free, 90 on Pro.

Why. [R] every save is a new immutable key. [M] Notion, Craft and AFFiNE give 7 days free and 30 paid, so 90 is a visible reason to pay.

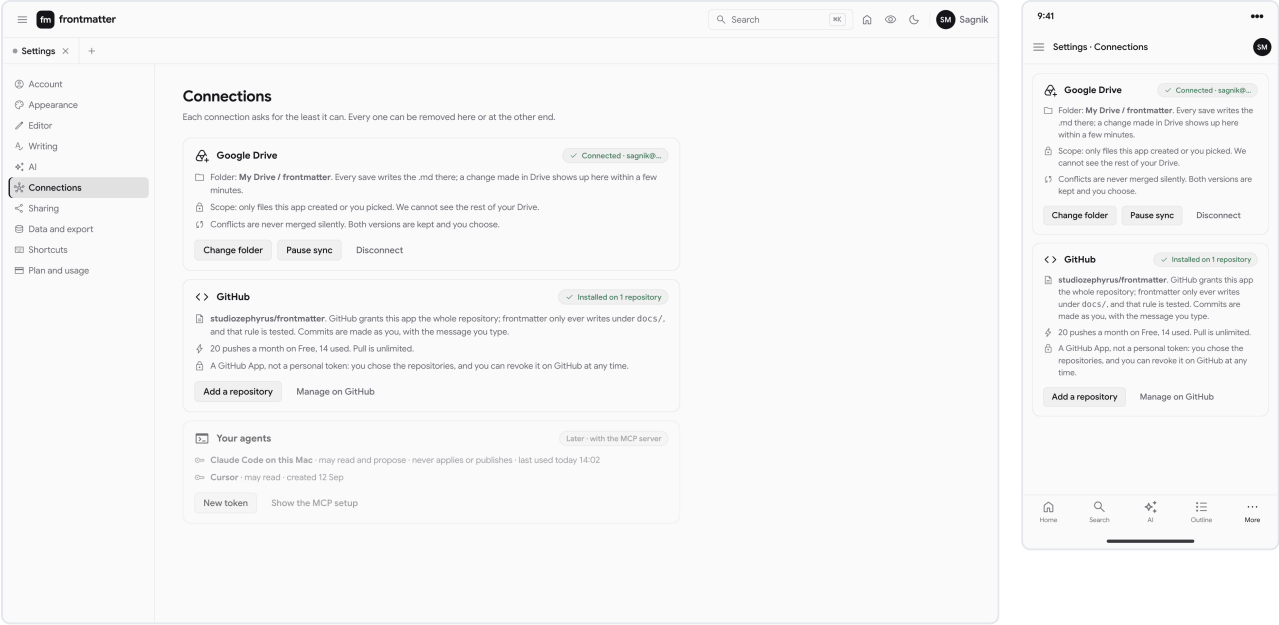
S22. Import



- Drop files or a folder. A folder keeps its structure and becomes a project.
- Six sources: a folder, GitHub, Google Drive, Google Docs, Word, a Notion export. A Google Doc over 10 MB is refused with the reason (F024).
- The progress panel says what was kept byte for byte, what was uploaded, and what needs a look. On the phone, sharing from another app is offered after install on Android and said plainly to be absent on iOS (F036).

Why. [Z] file and folder upload. [M] folder input works everywhere since Safari 11.1 and iOS 18.4. [M] Word converts in the browser with mammoth, so nothing is uploaded.

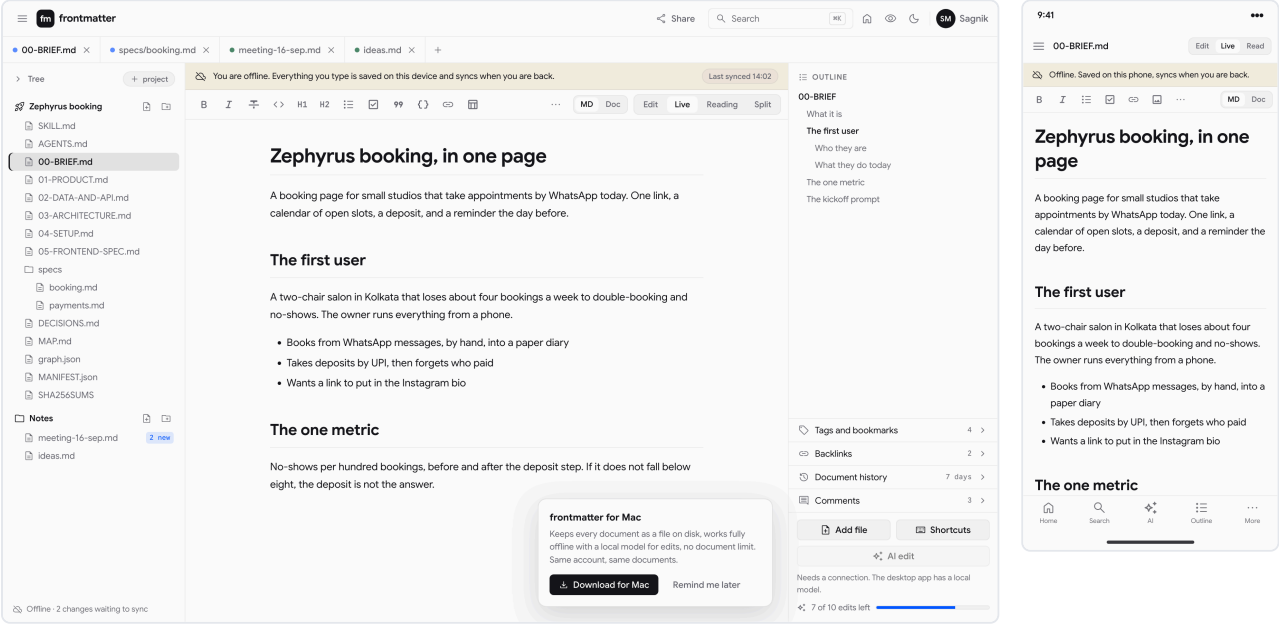
S23. Connections



- Google Drive: the folder, the scope, the conflict rule, "a change in Drive shows up here within a few minutes" (F027). Change folder, pause, disconnect.
- GitHub: "GitHub grants this app the whole repository; frontmatter only ever writes under docs/" (F034). Pushes used, revocable on GitHub.
- Your agents: marked Later, with the MCP server (F035).

Why. [Z] connect Drive and GitHub with the user's authorisation. [M] GitHub Apps carry "narrow, specific permissions" and installation tokens "expire after 1 hour". [R] permissions attach to the operation.

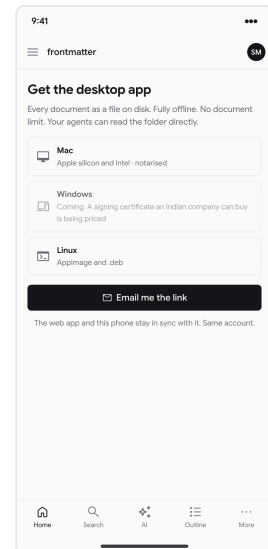
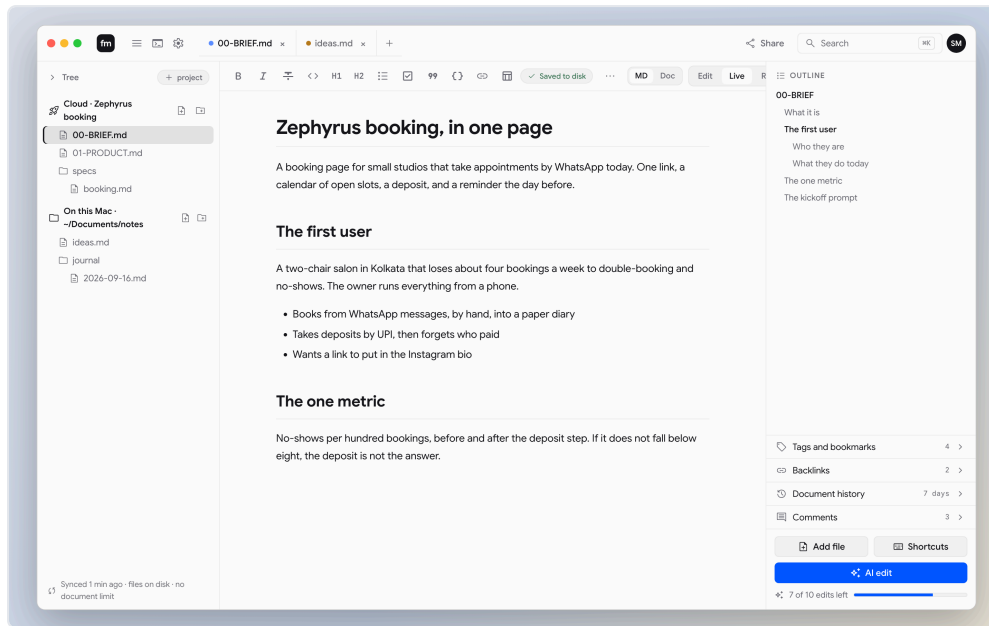
S24. Offline



- A banner, last synced time, changes waiting.
- AI edit is disabled with a one-line reason; on the desktop the local model takes over (F039).
- The desktop card: files on disk, fully offline, no document limit.

Why. [Z] offline in the browser with browser storage. [M] Safari deletes script-writable storage after seven days without a visit unless the app is installed, so the banner is honest and the desktop app is promoted.

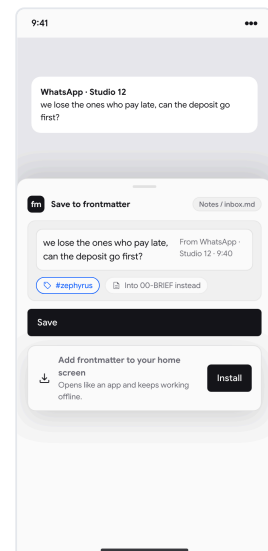
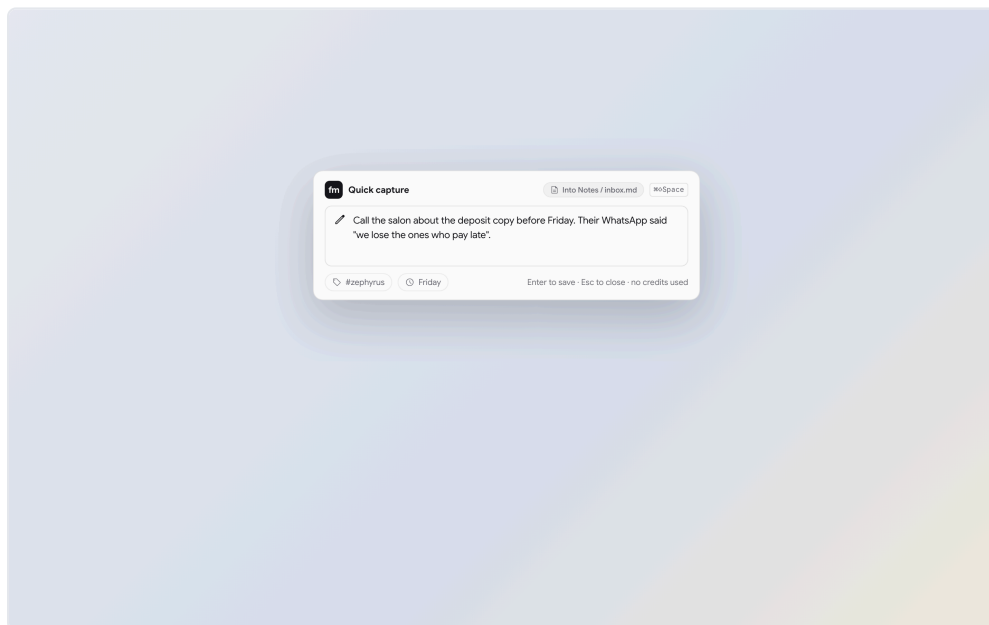
S25. Desktop app



- Cloud projects and folders on this Mac in one tree. Saved to disk.
- The same header, tabs and toolbar as the web.
- The phone page emails you the download link: Mac now, Linux unsigned, Windows coming (F047).

Why. [Z] the offline app is the one promoted. [M] Azure Artifact Signing's public trust is closed to organisations in India; Apple's programme is 99 USD a year; a commercial certificate for Windows is unpriced in this plan (F071).

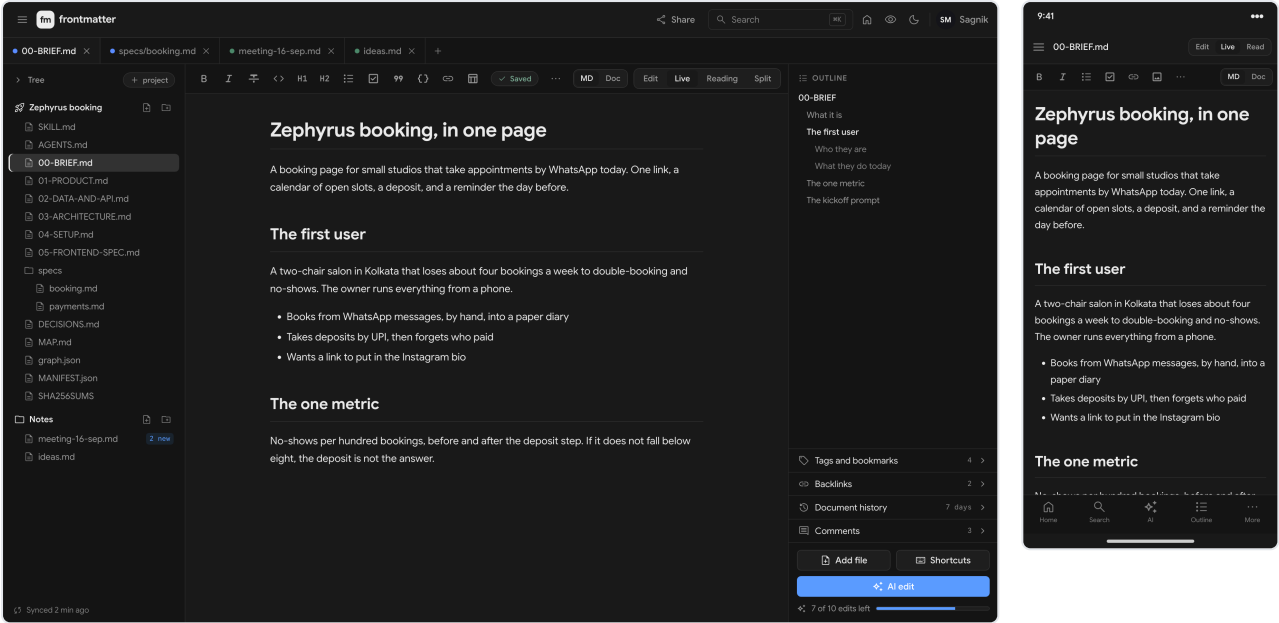
S26. Quick capture



- A global shortcut opens one box that saves into an inbox note. No credits.
- On the phone, the share sheet from any app lands in the same inbox, on Android after install.
- The install card appears once.

Why. [R] QuickAdd has 2,113,472 downloads, fast capture is named in 4 or 5 of the 9 note-taking methods we read. [M] the share target works only in installed Chromium apps, so the phone card asks to install.

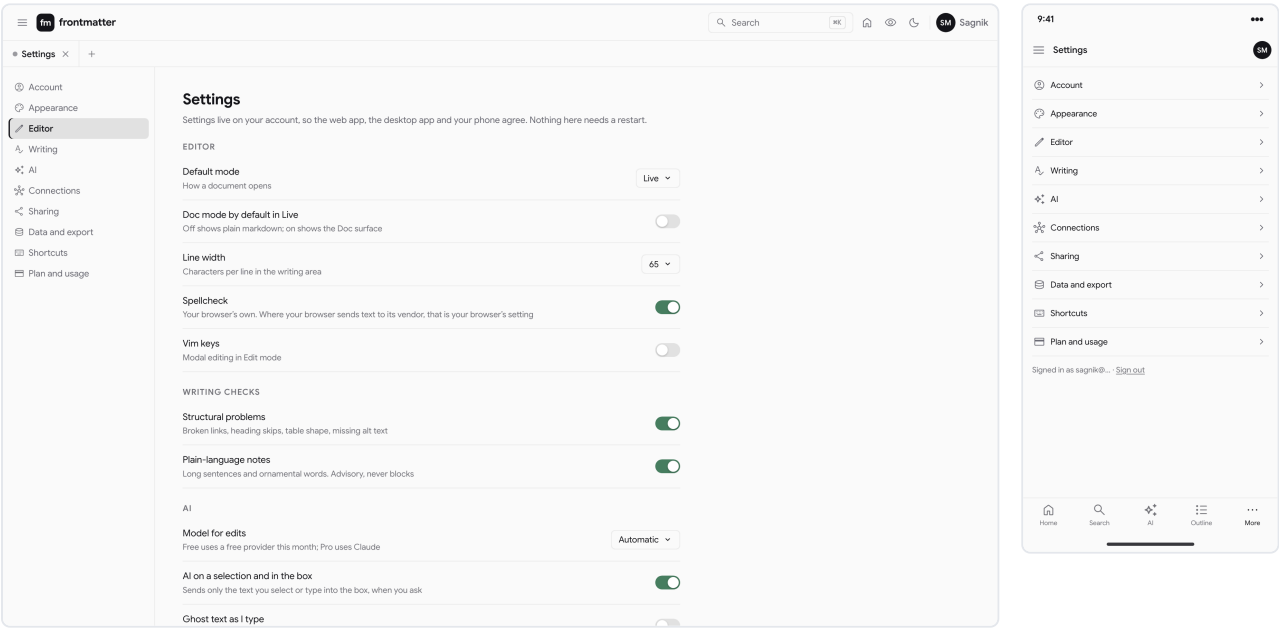
S27. Dark mode



- The same workspace on the dark tokens from globals.css.
- One toggle in the header, remembered on the account.

Why. [R] design praised 35 times in the reviews we counted, and dark mode is expected of every editor.

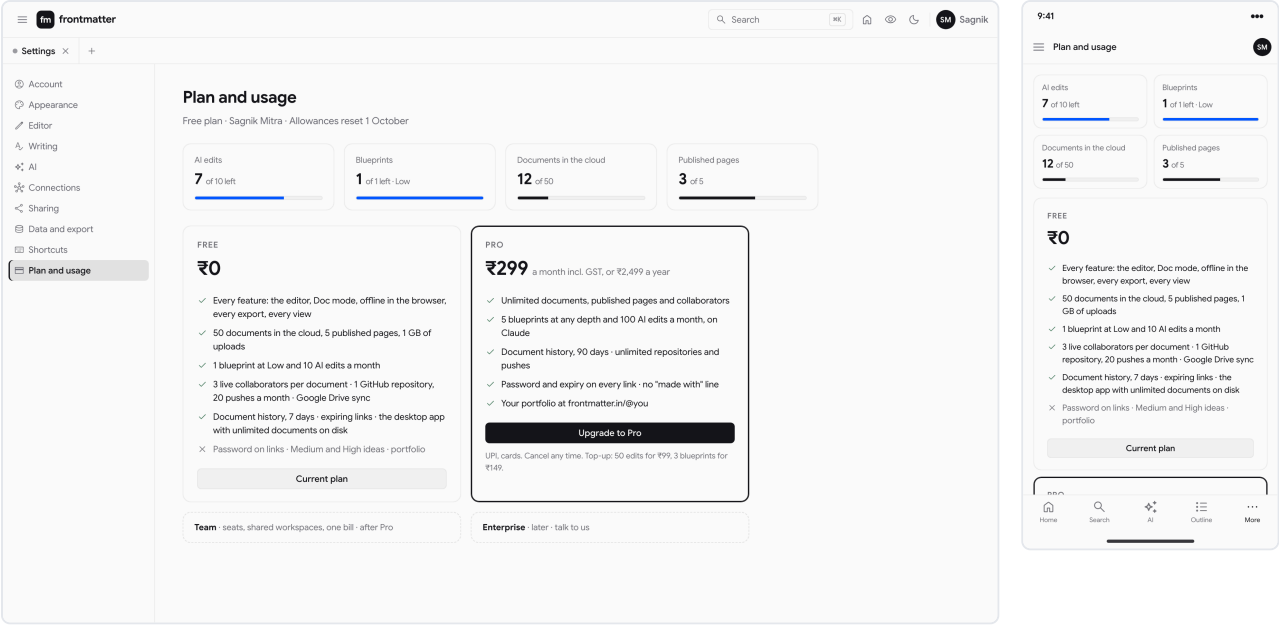
S28. Settings



- Ten sections, from Account and Appearance through Editor, Writing, AI, Connections and Sharing to Data and export, Shortcuts and Plan and usage.
- Spellcheck says it is the browser's own (F048). Two AI switches: ghost text as you type, off by default, and AI on a selection or in the box (F049). Mark AI text in the version record.
- Settings live on the account, so every device agrees.

Why. [Z] a proper settings page. [M] the sections are the union of Notion's and Obsidian's. [M] Obsidian's request for one settings set across vaults has 520 hearts, and ours is by design.

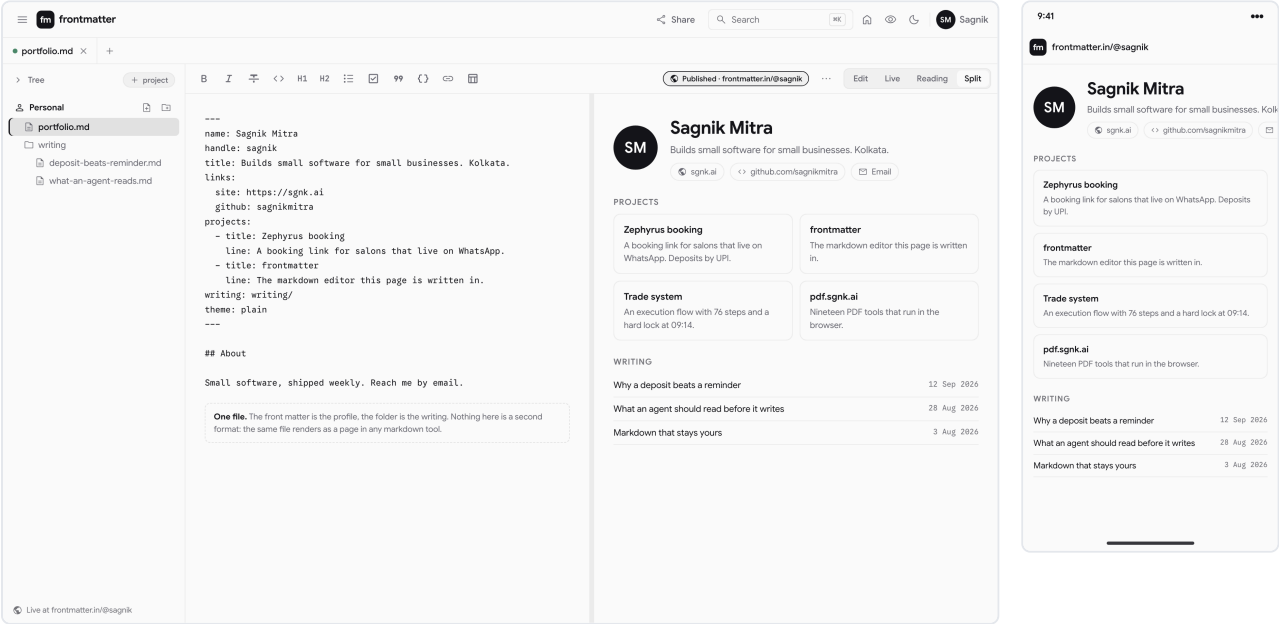
S29. Plan and usage



- Four meters: edits, blueprints, cloud documents, published pages. "Allowances reset 1 October" (F053).
- Free and Pro side by side, the price marked GST inclusive (F051). Team and Enterprise named as coming.
- UPI and cards, cancel any time, top-ups.

Why. [Z] Team and Enterprise are shown from day one. [L] Razorpay mandates are capped at ₹15,000 by RBI circular RBI/2022-23/73 of 16 June 2022, and Indian cards get one attempt (F074).

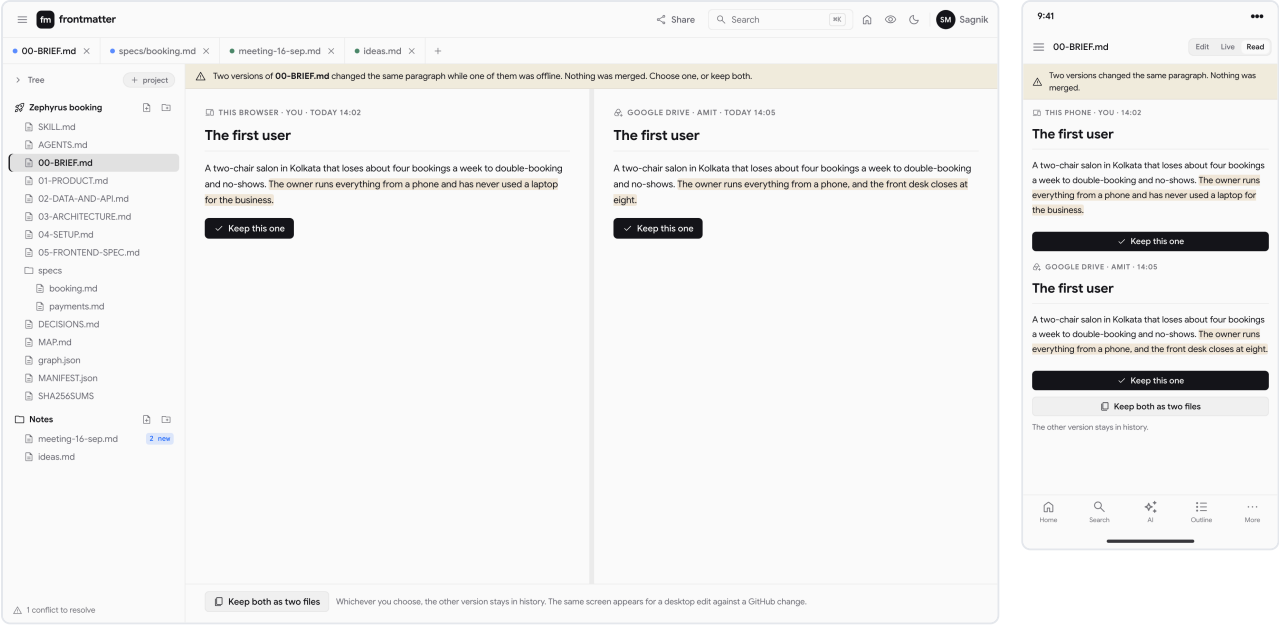
S30. Portfolio



- One file, portfolio.md. The front matter keys are name, handle, title, links, projects, writing and theme, the same list section 10 specifies (F052). The H2 sections are the page, the writing folder is the blog.
- Published at frontmatter.in/@handle, served from the stored file, no build step. No Follow button; there is no social feature in this plan.
- Pro, and late.

Why. [Z] a portfolio in Pro, like the reference site. [M] sayak.dev is Quarto markdown plus a config file and a build on GitHub Pages. Every generator opened wants a folder and a build. A one-file page is the gap.

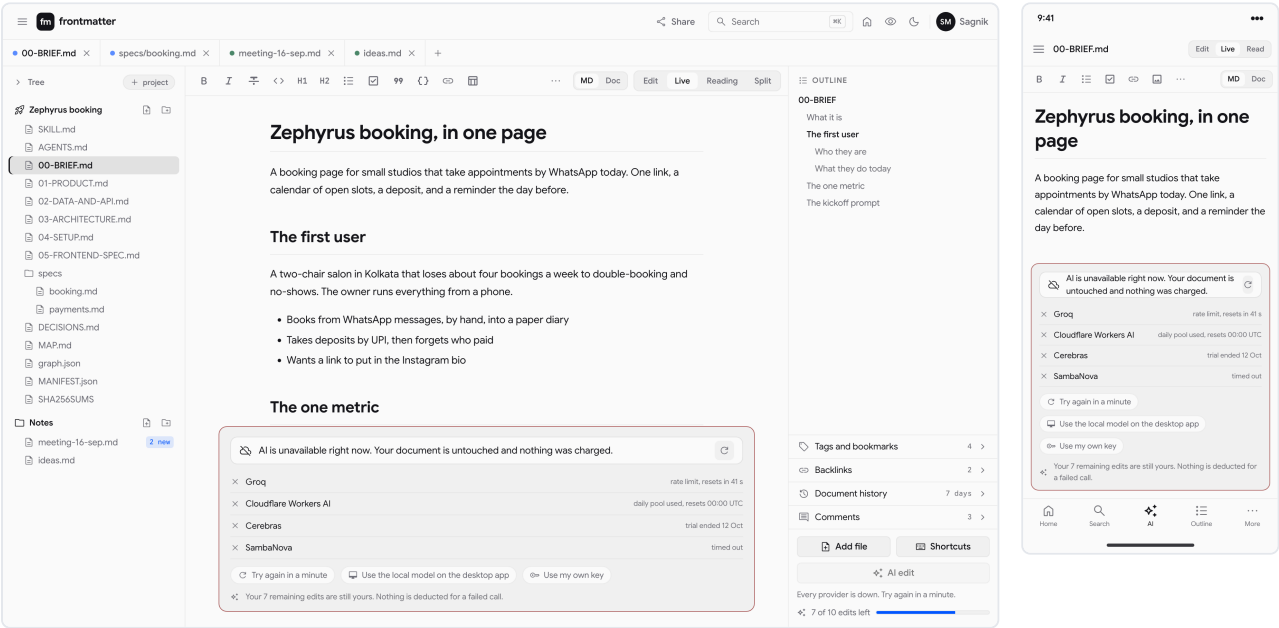
S31. Conflict



- Two versions of the same document side by side, each with its author, device and time. Nothing was merged.
- Keep left, keep right, or keep both as two files. The other version is always in history.
- The same screen serves a Drive edit against a web edit and a desktop edit against a GitHub change.

Why. [R] settled: no silent merge, ever. The audit found the plan's central promise had no screen.

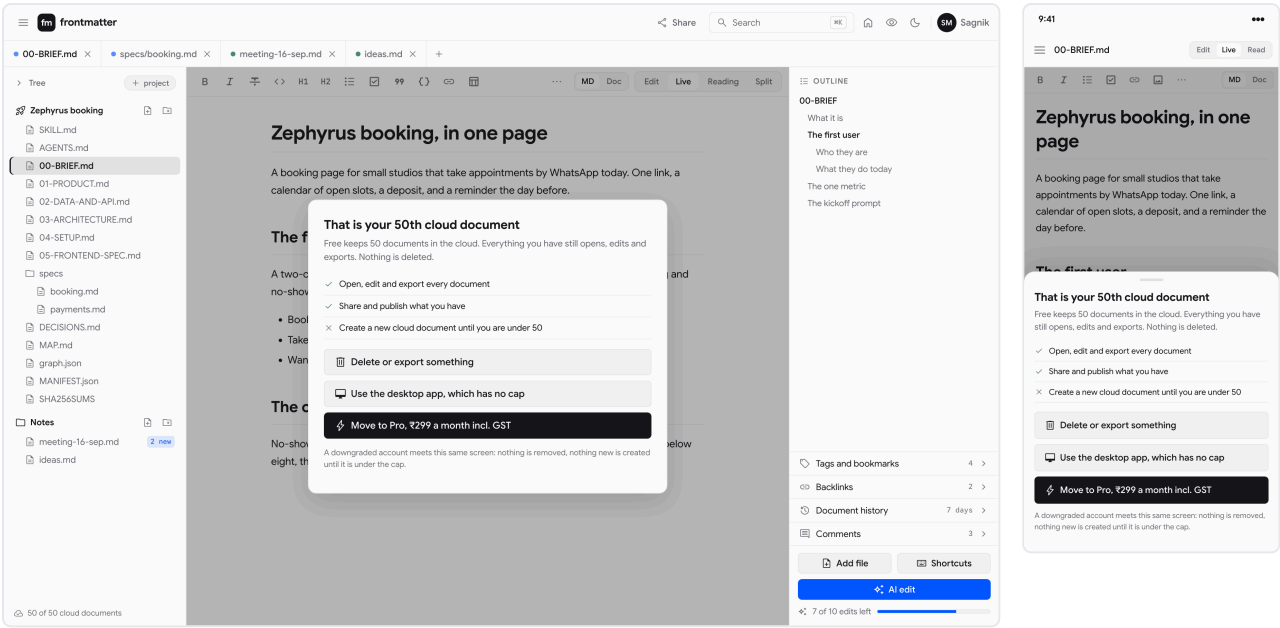
S32. AI unavailable



- The AI box when every provider in the chain has refused or timed out: the document is untouched, nothing was charged, try again in a minute, or on the desktop use the local model.
- The status of each provider in the chain, so the person knows it is not their document.

Why. [M] PAIR: "Provide paths forward from failure." [P] the chain has four links and the plan says what happens when the fourth fails.

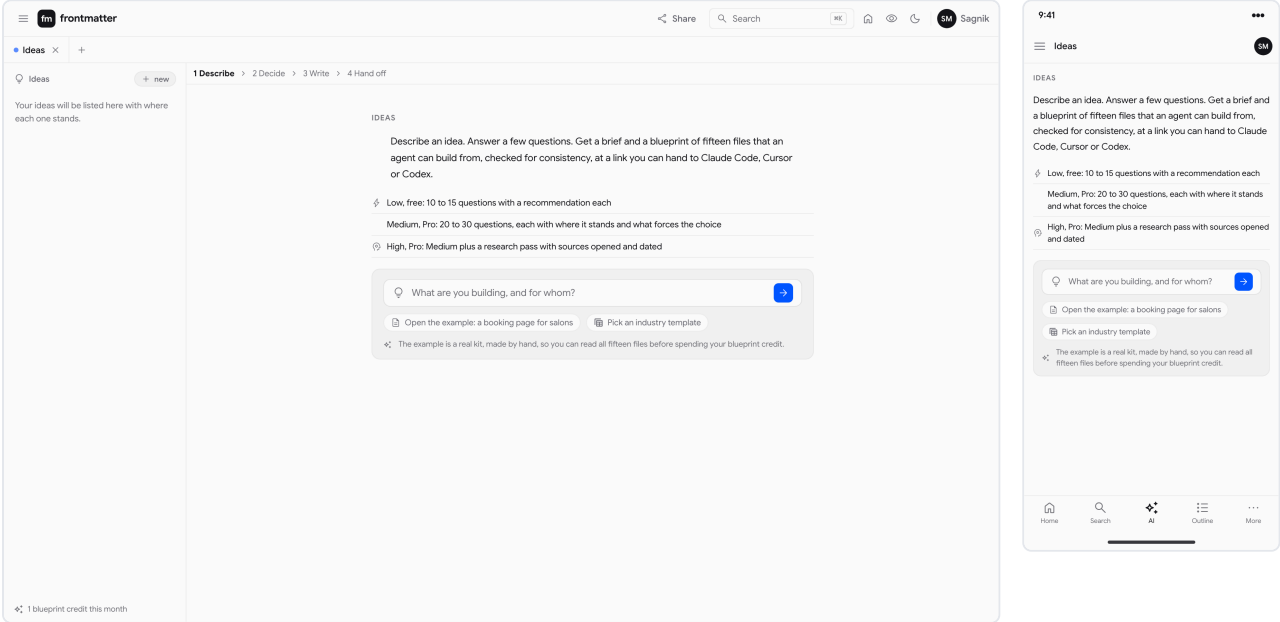
S33. Over the cap



- What happened: the 50th cloud document, or the tenth edit, or the fifth page. What still works: every document opens, edits and exports.
- What to do: delete or export something, wait for the reset, or move to Pro. The desktop app has no cap and is named.
- A downgraded account meets the same screen: nothing is deleted, nothing new is created until under the cap.

Why. [M] Nielsen: "Users often perform actions by mistake. They need a clearly marked "emergency exit"." [Z] founder question 9 on the caps and the downgrade.

S34. Ideas, empty

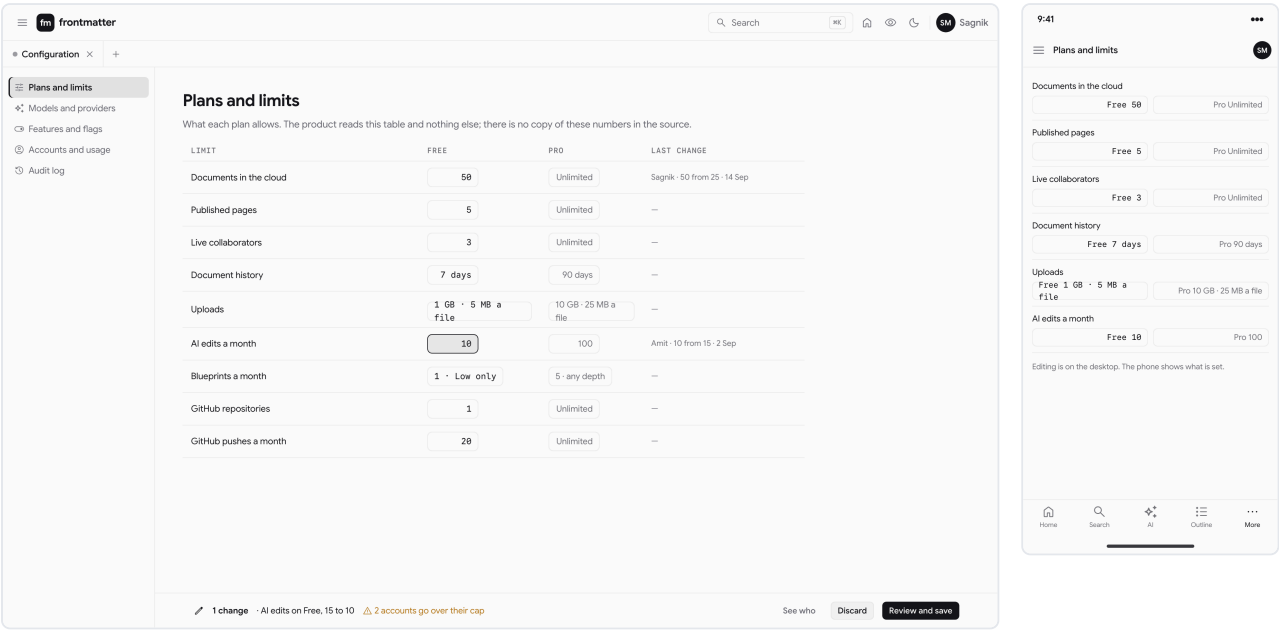


- The Ideas tab before the first idea: what a blueprint is in one line, the three depths in one line each, and one box to start.
- One hand-made example kit to open and read, so the person sees the fifteen files before spending a credit.

Why. [M] Nielsen: "Do not default to totally empty states." [R] the example kit is one of the twenty hand-made ones from Phase 0.

§7 5b. The configuration panel, which only a founder sees

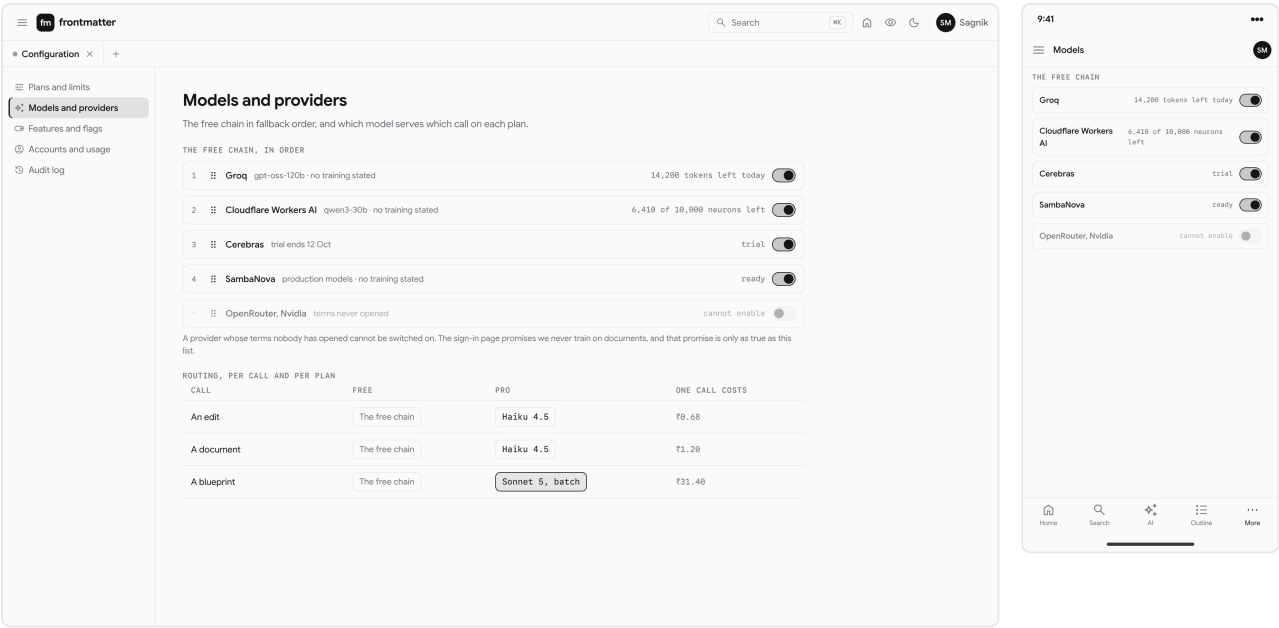
S35. Configuration, plans and limits



- Every limit in one table, Free against Pro, each cell editable. This row is what the product reads; there is no second copy in the source.
- Saving says how many accounts the change moves over their cap, and names them, before it writes.
- Each row carries its own last change: who, from what, to what, and when.

Why. [Z] the founders' decision of 17 September. [P] section 30 names one read path, `limitsFor(account)`, so a cap that appears anywhere else in the product is a defect the architecture gate should fail on.

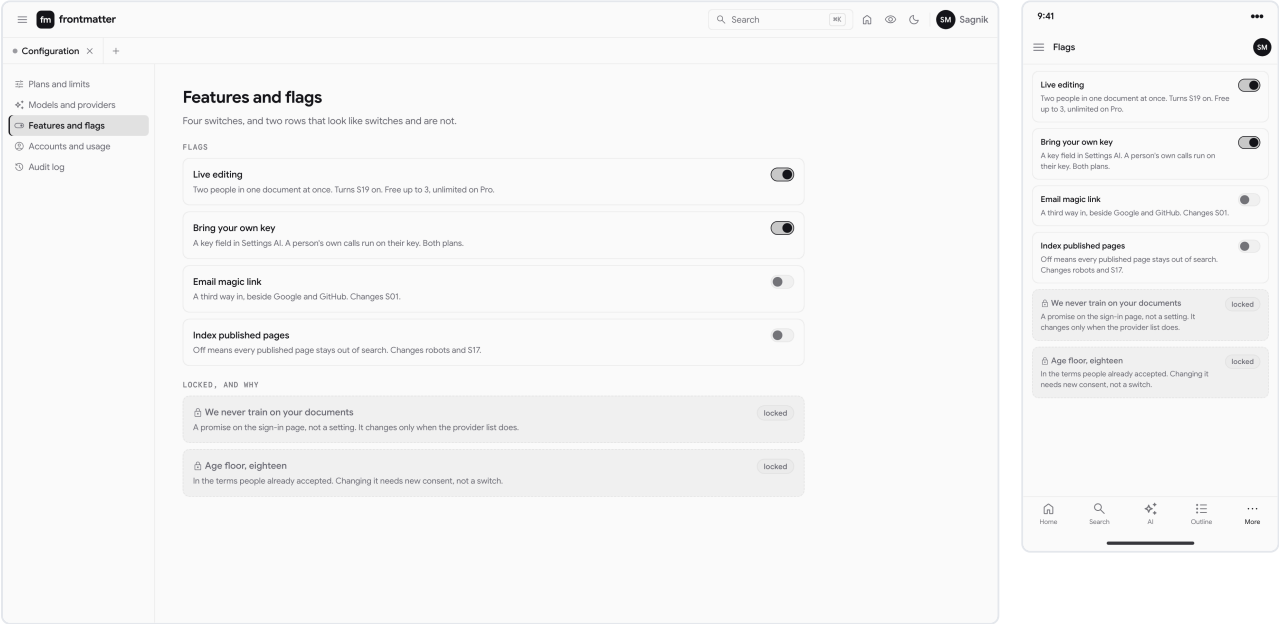
S36. Configuration, models and providers



- The free chain in fallback order, each provider on or off, with what is left of today's pool beside it.
- Routing per call type and per plan: an edit, a document, a blueprint, each naming its model and what one call costs.
- A provider whose terms nobody has opened cannot be switched on. The control is disabled and says so on the row.

Why. [Z] question 3 turns on keeping the sign-in promise true. [L] a provider list anyone can extend without reading terms is exactly how that promise breaks quietly.

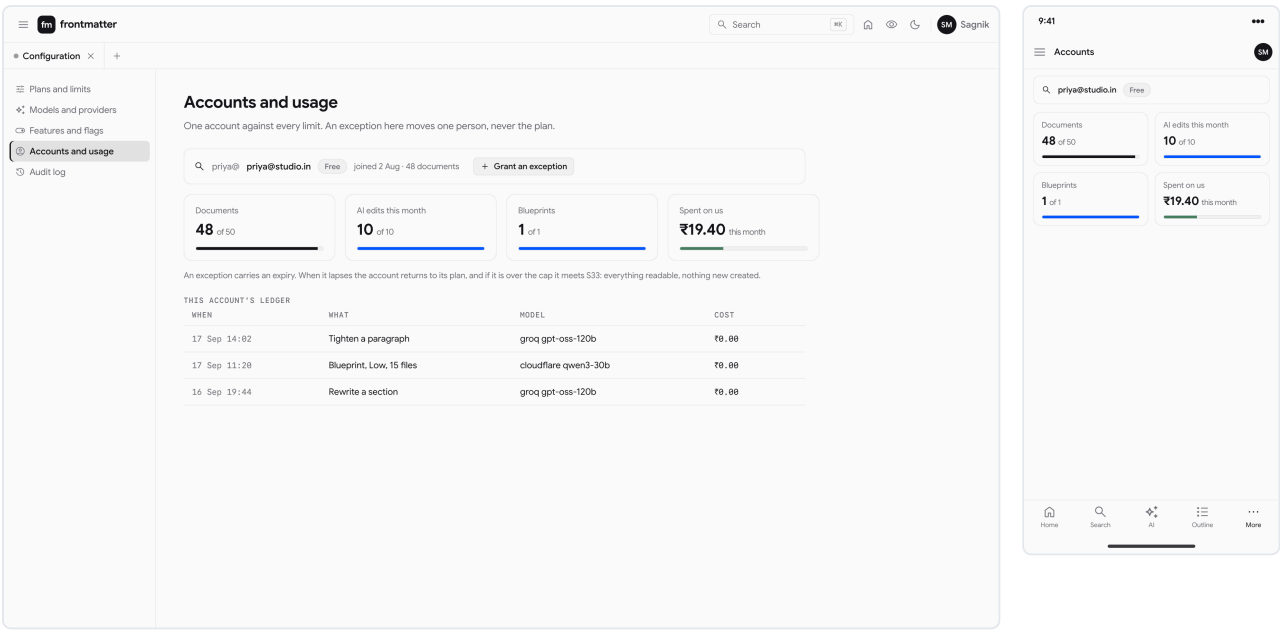
S37. Configuration, features and flags



- Four flags: live editing, bring-your-own key, the email magic link, and whether a published page is indexed by default.
- Each flag names the screens it turns on or off and the plans it reaches, so nobody has to guess what a switch does.
- Two rows are shown and locked: the training promise and the age floor. The reason sits on the row rather than in a document nobody opens.

Why. [Z] questions 8, 10, 17 and 18 become flags instead of decisions the build waits on. [L] section 30 says why the locked two cannot be flags: one is a promise, the other has already been consented to.

S38. Configuration, accounts and usage



- Find one account, see it against every limit, and grant a time-boxed exception without moving the plan for everyone else.
- That account's ledger: what it spent, on which model, and what it cost us.
- The audit log across every setting, newest first, read-only in the panel.

Why. [P] the ledger of section 18 is the counter every cap check reads, so this screen is a view of it rather than a second store. [Z] an exception is the honest answer to one support mail, and section 30 requires it to expire.

§6 Built in by default, so nobody needs a plugin

The rule: if a capability is a top add-on in three or more of the ecosystems we counted, it ships built in. Twenty pass [M].

Capability | Where it is a top add-on | Where it lives in frontmatter AI assistant and agent access | Docs, VS Code, Joplin, Logseq, Craft, Notion | The AI box, the AI menu; the agents card and the MCP server in Later Maths and LaTeX | Docs 52M+ installs, VS Code, Joplin, Typora | KaTeX in every mode Diagrams | VS Code 5.3M, Docs 47M+, Joplin, Logseq | Mermaid in every mode, Excalidraw as a block Tasks and to-do views | Joplin, Logseq, Craft, Notion, VS Code | Checklists, the tasks panel, the kanban view Templates | Joplin, Logseq, Craft, Notion 70,000+ | Home, the slash menu, the idea templates Editable tables | Joplin, Typora, Notion, Bear | Shipped today Outline and table of contents | Joplin, Typora, VS Code, Notion | The rail, a [toc] marker Export | VS Code 4.1M, Bear, Typora, iA | Markdown, HTML, Word, PDF, and the .md twin on every page Themes | Joplin, Typora, Bear, Logseq, VS Code | Appearance settings, dark mode Calendar and daily notes | Joplin, Logseq, Craft | The month panel, daily notes Kanban | Joplin, Notion, Obsidian 2,668,372 | View as Kanban Backlinks and link graph | Joplin, Logseq, Typora, Notion | Shipped today, plus the map Tags | Joplin, Bear, Craft, Logseq | Shipped today, rename and merge added Lint, style and spelling | VS Code 12.2M and 71.6M, Docs, iA | The problems panel, the formatter, the browser's spellcheck Image paste and resize | VS Code, Typora, Bear, Notion | Paste, resize, compress Note tabs | Joplin, Logseq, VS Code, Obsidian | Shipped today Slides | VS Code, Joplin, Obsidian | View as Slides Drawing | Joplin, Craft, Bear, Obsidian 7,974,073 | The Excalidraw block Version history and diff | Joplin, Craft, Notion, Docs | Document history Citations | Logseq, Docs, VS Code | A DOI lookup in the slash menu

What the earlier plan did not mention, now placed [0] : a table of contents marker, footnotes, emoji input, link previews for external links, OCR search inside images and PDFs through Tesseract.js, a formatter, an accessibility check on the document, and a stated API rate limit. Meeting notes and a sandboxed HTML block are refused, the second because the plan bans third-party scripts on published pages.

Notion's block inventory as a checklist [M] . Notion's API enum lists 32 block types; Code is documented outside the enum. frontmatter covers 27 of the 32 in plain markdown or a shipped block: paragraph, headings 1 to 4, bulleted and numbered items, to-do, toggle as a details block, quote, callout, divider, table and rows, image, video and file as links, PDF as a link, bookmark and link preview as previews, equation, table of contents, breadcrumb from the tree, child page and child database as links, embed as a link, template as a template, plus code as a fence. Not covered on purpose: column list, column, synced block, transcription, and the unsupported type, because none survives a plain markdown reader (F011).

Obsidian's plugins. The parity table of revision 3 stands: 15 plugin capabilities already ship, 21 are built in MVP 0, 8 in MVP 1. The Excalidraw block, templates, tasks, calendar, quick capture, ranked search and the importer are the big six.

§7 Doc mode, the honest scope

The research classified 64 Google Docs features, one row each, from Google's own help pages [M] . Twenty are plain markdown, fifteen need an extension, twenty-nine cannot live in a text file. The table is the specification a contractor builds from (F002).

Feature | Column | Carrier or reason Bold, italic | N | CommonMark Strikethrough | N | GFM Text capitalisation | N | Rewrites the letters Headings 1 to 6 | N | CommonMark ATX headings Bulleted, numbered, nested lists | N | CommonMark Checklist | N | GFM task list items Table, column alignment, sort rows | N | GFM tables Image, in line | N | CommonMark Emojis, special characters | N | Unicode text Links | N | CommonMark Outline, rulers, non-printing characters, zoom, dark theme, RTL view, shortcuts, copy and paste | N | Editor chrome, no file impact The @ menu | N | A command palette; what it inserts is classified per row Custom building blocks | N | Snippets of ordinary markdown Spelling and grammar, personal dictionary, autocorrect, Smart Compose | N | Tools over text Find and replace, word count | N | Tools Translate, voice typing | N | Tools; output is text Screen reader, braille | N | A property of the editor Offline | N | A local file is offline by nature Templates | N | A markdown file Download formats | N | A tool Underline | E | Pandoc [text]{.underline} Superscript, subscript | E | Pandoc ^ and ~ Font family, font size | E | A raw HTML span; Google drops it on export Text colour, highlight | E | Colour a raw span; highlight ==text== Title, subtitle | E | Frontmatter Alignment | E | Raw HTML or a span attribute; removed by Google on export Paragraph borders and shading | E | A Pandoc fenced div; plain elsewhere Column widths | E | Pandoc multiline tables Resize image | E | {width=} attribute Equations | E | \$math\$ Footnotes | E | [^1] , rendered by GitHub, Obsidian and Typora, not in the GFM spec Table of contents | E | A [toc] marker Bookmarks, internal links | E | A heading id and [text](#id) Variable chips | E | Frontmatter keys and a placeholder substituted at export Citations | E | Pandoc's citation syntax Line and paragraph spacing | X | A theme property Indentation, tab stops | X | Four leading spaces are code Custom bullet glyphs | X | List markers are fixed syntax Merge cells | X | Neither GFM nor Pandoc has them Cell background, borders, padding | X | No carrier in any specification Pinned header rows | X | Pagination Wrap, break, behind, in front of text | X | Google itself has none in pageless Crop and adjust | X | A pixel edit; save the file Drawings | X | A Google Drawings object; embed an image Linked charts, tables, slides | X | Live links to Sheets Page numbers, headers, footers | X | Pagination; front matter for export Page break | X | An export marker only Section breaks | X | Pagination Margins, page setup, orientation | X | Frontmatter for export Columns | X | Google itself has none in pageless Pages or pageless | X | Markdown is pageless Watermarks | X | Pagination Line numbers | X | A view Smart chips | X | Live data; Google's rule: "Smart chips change to text or links" Dropdowns | X | Live state Building blocks | X | Gmail and Calendar integrations AppSheet and third-party chips | X | External state Document tabs | X | One file per tab, or H1 sections Comments, action items, reactions | X | The

version record and the change queue Suggesting mode | X | Editor state; a proposal in the queue Version history | X | Document history Assign tasks | X | Google Tasks; the checklist item itself is N eSignature, Meet, Keep, add-ons | X | External services Publish to web | X | Hosting state

The substitutes for the twenty-nine [P] : a page-break marker and page setup as front matter honoured only by export; comments and suggestions in the change queue and the version record; versions in history; tabs as files.

The refusal rule [P] . Doc mode carries a feature only if a stranger's plain markdown parser, with no editor and no export theme, still shows the author's meaning from the bytes on disk.

What the rich editors admit [M] . Tiptap's markdown extension is "a early release" and says "Comments are not supported yet". Typora: "Custom fonts in Typora are set by CSS." Google's markdown export: "Font colors, highlights, and text alignment are removed." We are not inventing a limit. We are naming the one every tool has.

§8 What a document can become

Representation | Renderer | Licence | Signal | Ships Flow view | Ours: H2 as phase, H3 as step, bracketed tag, trailing reference | Ours | [Z] the founders' flow site | MVP 0 Diagrams | Mermaid, 90,268 stars, 12,122,962 weekly downloads, block, architecture and kanban types now exist | MIT | Renders on GitHub, Obsidian and HackMD | Ships today Drawing and canvas | Excalidraw, 132,141 stars, saved beside the note as JSON Canvas 1.0 | MIT | 7,974,073 plugin downloads | MVP 0 Slides | Marp core, splits on a horizontal rule, no new syntax | MIT | Slides from markdown: Advanced Slides at 837,129 Obsidian downloads renders with reveal.js; Marp's own signal is marp-vscode at 857,640 installs (F061) | MVP 0 Mind map | markmap, reads the outline, no new syntax | MIT | 885,474 plugin downloads | MVP 0 Kanban | Ours: headings as columns, task items as cards, the obsidian-kanban shape | Ours | 2,668,372 downloads, no open renderer | MVP 1 Charts from a table | Ours: an `fm-chart` block that points at the table above it | Ours | 324,208 downloads | MVP 1 Portfolio | Ours: one file with front matter at frontmatter.in/@handle | Ours | No comparable | Late, Pro Maths | KaTeX, 18,887,573 weekly downloads | MIT | Universal | Ships today Music | abcjs | MIT | HackMD renders it | Later PDF and print | Paged.js for the browser, Pandoc on the server | MIT, GPL | Universal | MVP 0

Two things we will not embed [M] : tldraw, whose licence forbids production use without a key and phones home, and D2, which is MPL and duplicates Mermaid. Database views over front matter, as Obsidian's Bases, have no open renderer and wait. The kanban and chart blocks copy the shape of obsidian-kanban and obsidian-charts and none of their code, because the first is GPL and the second AGPL, and both are unmaintained (F063).

§9 Idea mode, three depths

Depth | Who | Questions | What each answer carries | What it costs us Low | Free | 10 to 15 | A recommendation and one line of reason; Not sure recorded as open | About fifteen model calls per blueprint on the free chain Medium | Pro | 20 to 30 | Where it stands, what forces the choice, options with gains and costs, evidence from the person's own documents and the template's sources, dated when the template last checked them | The same calls on Sonnet through the batch API, plus a retrieval pass over the project High | Pro, 3 blueprint credits | 20 to 30 | Medium, plus a research pass before the questions: sources opened, dated and quoted, a decision record you can publish | A background job of about an hour, with web fetches; about 150,000 tokens in and 40,000 out on Sonnet, assumed until measured

The blueprint [Z] [P] . Fifteen files in a skill folder: SKILL.md, AGENTS.md, 00-BRIEF.md, 01-PRODUCT.md, 02-DATA-AND-API.md, 03-ARCHITECTURE.md, 04-SETUP.md, 05-FRONTEND-SPEC.md, specs/booking.md, specs/payments.md, DECISIONS.md, MAP.md, graph.json, MANIFEST.json, SHA256SUMS. Twelve are markdown documents, three are data files. The frontend spec is the founders' ask 4 and the file S12 promises (F025). Its cost model, assumed until measured (F017): one call per file at 5,455 tokens in and 2,091 out, so 81,825 in and 31,365 out per blueprint.

Templates [Z]. Seven industry templates ship: local service business, SaaS, marketplace, internal tool, mobile app, content site, agency. Each carries its question bank, its comparables and its sources. Generate one for my industry writes a new template from the idea and marks it as generated. Section 26 costs the writing.

What a template holds [R]. The question order, the recommendation rules, the file list, the consistency checks, and the sources a Medium answer may cite. Templates are markdown files in a folder, so a person can read, edit and share them.

The rule that keeps it honest [P]. Low never shows evidence it did not read. Medium cites only the person's documents and the template, and says when the template last checked a source. High cites only pages it opened, with the date. A recommendation is never a percentage [M] PAIR: do not show confidence when "The confidence level isn't impactful". Not sure is never written as the founder's decision.

§10 Sharing, publishing, portfolio

The four ways out. People with a role from section 19. A link with read or edit, an expiry, and on Pro a password. A published page at frontmatter.in/p/slug with a `.md` twin, a Made with frontmatter line on Free, and Report, Privacy and Terms in its footer. The portfolio at frontmatter.in/@handle on Pro.

Password links [M]. Dropbox: "add a password to a shared link" on Professional and above, not on Basic or Plus. Figma: "Available on all paid plans". Loom: "Business, Business + AI, or Enterprise". Notion: "Can I password protect a page? Unfortunately, not at the moment." We store a hash, ask once per browser, and put it on Pro.

Expiring links [M]. Bitwarden gives every user a deletion date with a default of seven days. Google Drive allows expiry only on Workspace editions and never on anyone-with-the-link. Ours is free, with a seven-day default and no upper bound.

Published pages [M]. No product opened caps public pages at a small number. Notion says "Unlimited published pages" and gates one custom domain. The market's gate is the domain and the branding. Ours is five pages on Free with the line, unlimited and unbranded on Pro, and a custom domain later. Pages are not indexed today; whether they should be is an open question in section 29, because the audit is right that not indexing removes search as a channel.

Portfolio [M]. sayak.dev is a Quarto project: one `.qmd` per page, a `_quarto.yml`, a build, GitHub Pages. Astro wants a content folder and a schema. We serve the stored file. The front matter keys are name, handle, title, links, projects, writing and theme, all ignorable by any other tool, so the file stays portable. Section 20 specifies them.

§11 Bringing things in and out

Folder upload [M]. The folder input works in Chrome 7, Edge 13, Firefox 50, Safari 11.1, iOS Safari 18.4 and Android Chrome 132. Drag-and-drop of a folder works on desktop through the entries API. Writing back to the person's own folder works only in Chrome and Edge through the directory picker, and MDN marks that API as not baseline. So: every browser imports, Chromium browsers can keep a live folder, and everyone else gets Drive, GitHub or a download for the way back. An Obsidian vault imports as it is. Its `.obsidian` folder is read for the daily-note path and the templates folder and nothing else.

Google Drive [M]. The `drive.file` scope covers files the app created or the person picked, and needs only basic verification. `changes.list` and `files.watch` accept it. A change channel lasts a week at most, with no automatic renewal, and carries no content, so the app polls the change list from a stored page token. The cost, re-derived with the poll included (F027): 30 saves a day at 50 units is 1,500 units; a poll every five minutes at 100 units is $288 \times 100 = 28,800$; 30,300 units per user per day. The daily project threshold of 400,000,000 units covers $400,000,000 / 30,300$

= 13,201 connected users before Google's quota increase, which the limits page now says is billed. A one-minute poll would cost 145,500 units a day and serve 2,749 users, so S23 promises "within a few minutes". Conflicts are never merged silently. Both versions are kept and the person chooses on S31.

GitHub [M]. A GitHub App, not a personal token. It asks for the Contents permission, read and write, and nothing else; GitHub grants that for the whole repository, so the promise to write only under docs/ is frontmatter's own rule, enforced server-side and tested (F034). Installation tokens expire after an hour and carry their own 5,000 requests an hour. Every update sends the file's blob sha and treats a 409 as a re-read, which is the splice engine's compare-and-swap rule in GitHub's words. Free: one repository and 20 pushes a month. Pro: unlimited.

Google Docs and Word [M]. Drive exports a Google Doc as `text/markdown`, and "Exported content is limited to 10 MB", which S22 states when it refuses (F024). Word converts in the browser with mammoth, so the file never leaves the machine. Google's own loss statement applies to both: colours, highlights and alignment are removed.

Notion [M]. The export zip imports as a project. The API allows 180 requests a minute on a non-business workspace, which is enough to pull a workspace page by page as a later feature.

The twenty public APIs worth using [M]. Seven run entirely in the browser with no key and send no text anywhere: KaTeX, Mermaid, Tesseract.js for OCR, pdf.js and pdf-lib, the DiceBear library, and self-hosted Google Fonts. Key-less and remote, sending only the query: Wikipedia and Wiktionary at 200 requests a minute with a user agent, the Free Dictionary, Datamuse at 100,000 requests a day until 1 January 2027, after which a key is mandatory, and customer-facing use needs prior contact, Crossref and OpenAlex for a DOI, Open Library for an ISBN, arXiv at one request every three seconds, Frankfurter for currency. Behind our proxy because they need a secret: Unsplash at 50 an hour in demo and 1,000 after approval with attribution and a download ping, Pexels at 200 an hour and 20,000 a month, DeepL's developer plan at a million characters, iframely at 2,000 hits a month, billed once an hour per URL. Never sent document text: LanguageTool's public endpoint, which says "Do not send automated requests" and caps a request at 20 KB and an address at 20 requests and 75 KB a minute (F062). Grammar beyond spelling waits for a self-hosted LanguageTool. Semantic Scholar receives DOIs and titles only, as good practice; its licence page carries no clause about training on what is sent, and the earlier sentence that said so is withdrawn (F062).

§12 Offline, desktop, phone

In the browser [M]. Every keystroke goes to IndexedDB or the origin private file system, which has been baseline since March 2023. Chrome allows an origin about 60 percent of the disk. Firefox allows the smaller of 10 percent or 10 GiB. Safari allows about 60 percent since macOS 14 and iOS 17, and deletes all script-writable storage after seven days of Safari use without a visit, unless the app is on the home screen. Background sync and the share target exist only in Chromium. The one rule: never let the browser be the only copy. Persist is requested inside a user gesture, and the first connection pushes everything to the server.

The desktop app [Z] [M]. Files on disk, no document limit, fully offline, agents can read the folder. It is the one we promote, and the web app stays. Each target builds on its own runner in CI: Tauri says cross-compiling Windows from macOS is "a last resort" and signing it needs an external tool (F072). Signing: macOS on the Apple programme at 99 USD a year; Linux unsigned by choice; Windows shown as coming until a commercial certificate is priced, because Azure Artifact Signing's public trust is closed to organisations in India (F071).

The phone [M]. Material: under 600 dp use a navigation bar with three to five destinations and one pane, confirmed on developer.android.com. Ours has five: Home, Search, AI, Outline, More. The tree and the right pane are drawers, as the shipped code does. Targets are large and spaced, which is Fitts's law; the number is Material's, not Fitts's (F015).

§13 Free and Pro

The founders' candidates and the market [M]. Forty pricing pages opened on 17 September; Craft, Docmost and the Obsidian roadmap re-opened on the 17th by this revision.

Cap | Founders' candidate | Market | Recommendation Cloud documents on Free | 5 | Unlimited at Google Docs, Notion for one person, Obsidian, HackMD, Bear and 15 others. Metered by size at Craft (1,500 blocks, 1 GB), Anytype (100 MB), AFFiNE (10 GB), Nuclino (2 GB). Counted at 50 by Nuclino, Evernote and UpNote. Figma's 3 files is the only lower number, and it comes with unlimited drafts | 50, and 1 GB of uploads at 5 MB a file. A blueprint alone is 15 files and an imported vault is hundreds, so 5 would block both funnels on the first day Published pages on Free | 2 | No product caps public pages at a small number. Notion: "Unlimited published pages". The market gates the custom domain and the branding | 5 with the Made with line. Pro unbranded and unlimited Live collaborators on Free | 1 | HackMD "3 invitees", AFFiNE "Up to 3 members per Workspace", Notion 10 guests, Confluence 10 users | 3 people per document History on Free | none | Notion, Craft and AFFiNE 7 days free and 30 paid | 7 days free, 90 on Pro Uploads | none stated | Obsidian Sync Standard 1 GB and 5 MB a file, Plus 10 GB | Free 1 GB at 5 MB a file; Pro 10 GB at 25 MB a file; a 30-day trash on both (F026) AI on Free | credits | Every free tier with AI puts a number on it: Kiro 50 credits, Tana 50 queries, Mem 25 messages, Canva 20 uses, Craft 15 credits, GitBook 10 messages a week, ChatPRD 3 chats | 10 edits and 1 Low blueprint a month, top-ups on Pro GitHub on Free | open question | HackMD: free with "20 GitHub pushes per month", unlimited at \$5. GitBook: Git Sync free. Notion gates GitHub to Business at \$20 | Free with 20 pushes and 1 repository. Pro unlimited. The connection is the product, so it cannot sit behind a higher tier Password links | asked for | Paid at Dropbox, Figma and Loom. None at Notion | Pro Google Drive | asked for | Notion gates Drive to Plus at \$10. It is the person's own storage and costs us nothing | Free Downgrade | not stated | Figma keeps files readable over the cap | Every document stays readable and exportable; nothing new is created until under the cap (S33)

Pricing [Z] [M]. Free at ₹0. Pro at ₹299 a month or ₹2,499 a year, both GST inclusive, which is ₹253.39 before GST and about ₹246 after Razorpay's fee, section 22. That is about \$3.12 at ₹95.96 to the dollar. The first paid tier across peers, corrected this revision (F004, F059): UpNote \$1.99, Bear \$2.99, Joplin Cloud 2.99 euro, Obsidian Sync \$4, Anytype \$4, HackMD \$5, Craft ₹526.7 a month on its India page, Confluence \$5.42, AFFiNE \$6.75, Standard Notes \$7.50, Nuclino \$8, Mem \$9, Capacities \$9.99, then Notion, Slite, Outline, Linear and Reflect at \$10; Docmost is \$6 a seat with a ten-seat minimum, so \$60 a month; Notesnook's India page, as the audit read it, is below ₹299, and it could not be re-opened by this revision. AI-forward tools sit at \$15 to \$20. So: under every collaboration and AI tool, above the single-user note apps, with more in the box than either. Top-ups: 50 edits for ₹99, 3 blueprints for ₹149. Team, with seats and one bill, after Pro. Enterprise later. Max, a tier above Pro with more credits, later and unpriced (F023). No student tier [Z].

What Pro buys, in one line [P]. Unlimited documents, pages and collaborators, 10 GB, 90-day history, password links, Medium and High ideas, 100 edits and 5 blueprints on Claude, the portfolio, no branding.

§14 AI for the pilot, and after

The three tasks [R] [P]. An edit on a selection is about 4,000 tokens in and 800 out. A document is 2,000 in and 1,500 out. A blueprint is fifteen calls, 81,825 in and 31,365 out in total, assumed until measured (F017).

Who allows a free pilot, verbatim from their pages [M].

Provider | Free models | Free limits | Trains on prompts | In the chain Groq | gpt-oss-120b, gpt-oss-20b, qwen3.8-27b, compound | 30 requests a minute, 1,000 a day, 8,000 tokens a minute, 200,000 a day, per organisation | No: "Groq is not permitted to use Inputs or Outputs for training" | Yes Cloudflare Workers AI | Any non-gated model, qwen3-30b among them | 10,000 neurons a day, 300 requests a minute | No: "Cloudflare does not use your Customer Content to (1) train any AI models made available on Workers AI or (2) improve any Cloudflare or third-party services" | Yes Cerebras | gpt-oss-120b, qwen-3.8-27b | A trial: "\$5 in free credits after adding a verified payment method. These

credits expire 30 days after they're granted"; 5 requests a minute, 1,000,000 tokens a day | No | While the trial lasts (F005) SambaNova | DeepSeek-V3.1, Llama 3.3 70B, gpt-oss-120b | 20 requests a day per model | No | Yes, production models only OpenRouter free endpoints | 24 of 444 models at zero, served by other providers | 20 a minute, 50 a day | "Each provider on OpenRouter has its own data handling policies"; the Nvidia-served endpoints' terms were not opened | No, until a provider's terms are opened and quoted (F006) Gemini API | Ten Flash and Pro models | Per-model numbers not public | Yes: "Google uses the content you submit to the Services and any generated responses to provide, improve, and develop Google products", and "Do not submit sensitive, confidential, or personal information to the Unpaid Services" | No Mistral Free | \$10 a month of credit | In the admin panel only | The free plan's training row is ticked with no opt-out | No GitHub Models | none | "fully retired" on 30 July 2026 | n/a | No Ollama on the desktop | llama3.2:3b at 2.0 GB, qwen3:4b at 2.5 GB, gemma3:4b at 3.3 GB, qwen3:8b at 5.2 GB fit an 8 GB machine | None | No: runs locally | Yes, desktop only

The routing [P]. Edits go to Groq gpt-oss-120b first, then Cloudflare qwen3-30b, then Cerebras while the trial lasts, then SambaNova. Documents go to Cloudflare first, then Groq. Blueprints go to Cerebras while the trial lasts, then Cloudflare, then paid Cloudflare neurons. The desktop app offers a local model for edits, with nothing leaving the machine. Never in the chain: Gemini's unpaid tier, Mistral Free, any OpenRouter endpoint, and anything whose terms were not opened. The founders confirm the chain's shape as question 3.

What the free pools carry, as one shared budget each, SIMULATED from the caps [0] (F007). At the caps a free user uses 10 edits and 1 blueprint a month. Per user per day: a third of an edit and a thirtieth of a blueprint. On Cloudflare, an edit costs 42.9 neurons and a blueprint 1,334 ($81,825 \times 4,625 / 1e6 + 31,365 \times 30,475 / 1e6$), so a user costs $14.3 + 44.5 = 58.8$ neurons a day and 10,000 neurons serve 170 users. On Groq, a user costs $1,600 + 3,773 = 5,373$ tokens a day and 200,000 tokens serve 37, though Groq's 8,000 tokens a minute make a blueprint a fourteen-minute job there. On Cerebras, 1,000,000 tokens a day serve 186 while the trial lasts. SambaNova's 20 requests a day are a smoke test. So the free chain carries about 200 active free users at the caps once the Cerebras trial ends, and about 390 while it lasts. Past that, the cheapest paid step is Cloudflare's overage at \$0.011 per 1,000 neurons: \$0.0194 (₹1.86) per free user per month at full caps.

Cost when the free pools are gone, 1,000 free users at their caps, per month, SIMULATED [M]. Input $1,000 \times (10 \times 4,000 + 81,825) = 121.8$ million tokens, output $1,000 \times (10 \times 800 + 31,365) = 39.4$ million. Cloudflare qwen3-30b at \$0.051 and \$0.335: \$6.21 + \$13.19 = \$19.40. Groq gpt-oss-20b at \$0.075 and \$0.30: \$9.14 + \$11.81 = \$20.95. gpt-oss-120b on Groq or Together at \$0.15 and \$0.60: \$18.27 + \$23.62 = \$41.89. Anthropic Haiku 4.5 at \$1 and \$5: \$121.83 + \$196.83 = \$318.66. Sonnet 5 at \$2 and \$10: \$243.65 + \$393.65 = \$637.30, or half with batch processing.

Pro's model cost, re-derived with the working shown, SIMULATED [0] (F001). At the Pro caps, 100 edits and 5 blueprints. Sonnet 5 for everything: an edit is $4,000 \times \$2 / 1e6 + 800 \times \$10 / 1e6 = \$0.016$, so 100 edits are \$1.60; a blueprint is $81,825 \times \$2 / 1e6 + 31,365 \times \$10 / 1e6 = \$0.164 + \$0.314 = \$0.477$, so 5 are \$2.39; the total is \$3.99, which is ₹383 at ₹95.96, against ₹246 net of GST and Razorpay. That loses ₹137 on every fully active Pro user, which is why revision 4's "about ₹120" was wrong and is withdrawn. The default routing now: Haiku 4.5 for edits at $4,000 \times \$1 / 1e6 + 800 \times \$5 / 1e6 = \$0.008$, so \$0.80 for 100; Sonnet 5 through the batch API for blueprints at half price, \$0.239 each, so \$1.19 for 5; the total is \$1.99, ₹191, leaving about ₹55 on a fully active Pro user before fixed costs. Haiku for everything is the same ₹191. Prompt caching is measured in week one and is not counted here. The price and the routing are founder question 2.

Work that never touches a model [P]. Structural checks, the formatter, search, the map, table-to-chart, word counts, spellcheck, OCR, dictionary, citations, currency, link previews, folder import and every representation. That is most of what the editor does in a day, and it costs nothing per use.

Security [R] [M]. We render untrusted markdown, run a model over private documents, and write to GitHub. A memory vendor said it this month: "Persistent memory makes prompt injection durable." Eight controls ship, the last two restored from revision 3 at the audit's finding (A21): no third-party scripts on published pages; agent tokens that may propose but never apply; a per-account budget and breaker on every model call; every model call attributed and

logged; documents sent to a model only when the person asks, with ghost text off by default; the change queue, so every agent change is read before it lands; document text reaches a model inside a delimited data block with a standing rule that content inside it is data; remote images in a shared document are proxied or click-to-load. And the kickoff prompt verifies the kit's tarball against a hash printed on the published page before it unpacks anything, and tells the agent to read before it builds (F028). Logs: an append-only log store in India, section 15 and section 23 (F069).

§15 The stack, costed

The question [Z]. Cloudflare R2 with a Firestore database: is it free to start, does it scale, and what is the perfect stack?

Three stacks at 1,000 users, per month, SIMULATED [M]. Assumptions: 10 documents of 20 KB each, 30 saves a day stored as full immutable copies, 100 MB of uploads each, 200 live sessions a day of an hour with two people, two developer seats on Vercel.

Stack | Month one | Where it stops being free | Hard limits R2 plus Firestore in Mumbai, Firebase Auth, Durable Objects, Vercel Pro | \$43.07 with WebSocket hibernation, \$85.57 without, on the research's Mumbai prices; about \$43.15 on the audit's | R2 at 100 users, Firestore at 333 users and then only on a Blaze billing account | 1 MiB per Firestore document, quotas reset at midnight Pacific; Mumbai storage \$0.165 a GiB-month by the research's reading of the page, \$0.104 by the audit's, and the page is a script shell to curl so neither can be reproduced (F040) R2 plus Supabase Pro in Mumbai for records, ledger and auth, Durable Objects, Vercel Pro | \$65.19 | Supabase Pro is \$25 from day one; its Free tier pauses "after 1 week of inactivity" and holds 1 GB of files | 8 GB disk then \$0.125 a GB, 100 GB of files included, 100,000 monthly active users Cloudflare only: R2, D1, Durable Objects, Workers | \$1.49 to \$6.49, plus \$40 if the app stays on Vercel | Workers Paid at \$5 for any splice over 10 ms of CPU | D1 is 10 GB per database and single-threaded, so tenants shard from the start, and Next.js would have to move to Workers

The verdict, and the decision [Z] (F016). R2 is right for the bytes on every stack: \$0.015 a GB-month, free egress, no minimum, and the 1,000-user workload never leaves its free operation tiers. On the database revision 5 recommended Supabase Pro, and on 17 September the founders decided otherwise. **The stack is the Next.js app we already run, Cloudflare R2 for bytes, and Firestore for records, with Firebase Auth for sign-in.** It is the cheaper row above, \$43.07 against \$65.19 at 1,000 users, and it is what the shipped code already initialises, so phase A stops being a migration.

The objections revision 5 raised against Firestore do not disappear. They are carried as build constraints, each with the place it is handled, and none of them is an argument to reopen the decision:

Objection | How the build handles it A Blaze billing account is required, and Google's FAQ links a page for when an Indian card is not accepted | Phase 0, with the company card, before any code depends on it A document cannot exceed 1 MiB | Document bytes live in R2 and never in Firestore; a Firestore document holds only metadata and a content hash, section 18 The free quota resets at midnight Pacific, not midnight in India | The caps the product enforces are ours, in the ledger, not Google's; Google's quota is a floor we stay well under, and the panel of section 30 can lower a limit if it is ever approached A credits ledger wants transactional writes | Ledger entries are append-only and never updated, so the balance is a sum over a collection rather than a row that two writers race for, section 18

So: the Next.js app on Vercel, R2 for bytes, Firestore for records and the ledger, Firebase Auth for sign-in, Durable Objects with the Hibernation API for live sessions. The first bill is \$20 for one Vercel seat, plus Firestore and R2 usage, which at 1,000 users is about \$2 and under \$1. It is not free to start, because Vercel writes: "Our Hobby plan is for personal, non-commercial use." A product that charges through Razorpay is commercial from its first rupee.

What the code runs today, and what phase A keeps [0] (F031). The shipped app signs in through GitHub with Auth.js and through Google with Firebase Auth, and initialises a Firestore client; a prototype `firestore.rules` designs document holding in Firestore. Under the founders' decision all of that stays. Phase A hardens `firestore.rules` from a

prototype into the rules the product runs on, keeps both sign-in paths, and migrates the local drafts the shipped app already holds under its legacy keys into the signed-in account (section 25, A33). The audit's proposal to delete `firestore.rules` (F045) is withdrawn: the file is live work now, not dead code. This is the one place where a founder decision made the build smaller rather than larger.

Live editing without reopening the settled decision [R] (F055). The document of record stays markdown bytes, one version per save. A live session is a shared document in a Durable Object that exists only while two people have the file open. Yjs holds that shared state and nothing else: every save writes a new version through the splice engine, the session's state is discarded when the last person leaves, and no CRDT state is ever persisted or merged into a file. That is the exception the record already allowed, and the written rebuttal to Zed Delta that the settled position owes is due before phase D. Whether to keep live editing at all is question 8.

Where the objects live [M] (F041). Durable Objects offer jurisdictions for the EU and the US only; the apac hint is "a best effort and not a guarantee", and an object does not move after creation. R2 likewise takes only an apac hint. So a session object is created from the first participant's request with the apac hint, its round trip from Mumbai is measured in the pilot, and the plan says plainly that the bytes sit under an apac hint, not in India.

Constraints the plan must respect whichever stack is chosen [M] : budget a Vercel seat from day one; use the Durable Object Hibernation API, because without it live sessions are the largest line at \$42.50 and the free plan fails at 28 sessions a day; send 100 MB uploads straight to R2 with a presigned URL, because Workers cap a request body at 100 MB; store deltas or deduplicate versions, because full-copy saves grow 18 GB a month per 1,000 users and that is the only line that compounds; keep an append-only security log for 180 days in Indian jurisdiction, which Sentry's and PostHog's free tiers may not pin to India, so the log store is its own line at R2 in Mumbai (F069).

The rest of the stack [M] . Auth through Firebase Auth for Google and Auth.js for GitHub, both already shipped, with an email magic link as the fallback sign-in the founders decide on (A50). Email through Resend, 3,000 a month free, for the 24-hour notices Razorpay mandates need. Errors through Sentry's free 5,000 a month. Analytics through PostHog's free million events. Search in the browser across the open workspace first, because Firestore has no full-text index, and Typesense when a server-side index is needed. Live editing through Yjs on Durable Objects, with Liveblocks as the fallback if we want a vendor, at 10 connections a room free.

§16 Principles, applied

Every rule below is quoted from the page it came from, and each names the screen it governs [M] .

- **Progressive disclosure, two levels at most.** Nielsen: "designs that go beyond 2 disclosure levels typically have low usability". The toolbar has its twelve buttons and one More. The AI box has four chips and one second row. Settings has ten sections and no sub-sections.
- **No tour.** Nielsen: "Tutorials interrupt users, don't necessarily improve task performance, and are quickly forgotten." Apple: "Consider providing a collection of context-specific tips instead of a single onboarding flow." A tip appears on the first hover of a control, once.
- **Empty states are pathways.** Nielsen: "Do not default to totally empty states." Home shows five starts. The review queue, when empty, says what would appear. The Ideas tab has S34.
- **Sign-in first, named honestly.** Apple: "Delay sign-in for as long as possible." Nielsen: "Users want to start using the product right away." Jakob's law is the one source that supports the founders' choice: "users prefer your site to work the same way as all the other sites they already know", if the reference class is Google Docs and Notion. It is. So the sign-in is one tap, the page shows the editor, and published pages need no account.
- **Reversible, visible, confirmable.** Nielsen's direct manipulation: "physical, incremental, and reversible actions whose effects are immediately visible on the screen." Every AI proposal previews in place, applies on a click, and undoes in one step.
- **No confidence numbers.** PAIR: do not show confidence when "The confidence level isn't impactful"; prefer alternatives, which "prompts the user to rely on their own judgement". Idea mode shows options and a recommendation, never a percentage.
- **An off switch.** PAIR: "allow users to adapt the output to their needs, edit it, or turn it off." Settings has it, and ghost text is off by default.
- **Speed.** Nielsen: 0.1 seconds feels instant, 1.0 second keeps the flow, 10 seconds needs a progress bar. Section 21 turns those into targets of our own.
- **Phone.** Material, read in a rendered browser on 17 September because the site is a script shell to curl: a navigation bar with three to five destinations under 600 dp, "Don't use navigation bars for desktop layouts", and the 600 dp compact breakpoint confirmed on developer.android.com (F043). Apple: "show no more than two levels of hierarchy in a sidebar", and do not hide it by default on desktop.
- **Accessibility.** WCAG 2.2 AA is the plan's target for every screen; the muted, danger and success tokens are retuned to 4.5:1 and the generator asserts it (F030). Whether the Rights of Persons with Disabilities Act 2016 section 46 reaches a private browser product turns on rule 15 of its rules, which is a counsel question in section 23 (F075). The draft amendment of 23 July 2026 would settle it the wide way: IS 17802 for every website, app or piece of software offered to persons in India, with a published conformance report. So the plan tests to IS 17802 alongside WCAG from the first screen rather than waiting for the rule to land.
- **Appetite, not estimate.** Shape Up: "Appetites start with a number and end with a design." Each phase in section 26 carries an appetite in weeks.
- **SOLID, as it applies here.** Martin: "Gather together the things that change for the same reasons." Each block kind is its own module. Adding a block or an AI verb is a registration, never an edit to the splicer. Every block honours one contract: parse a byte range, render, serialise byte-exact, report its splice range. Ports are small and client-specific. The engine depends on abstract ports for storage, the model and the change queue, and the R2, Firestore, GitHub and model adapters depend on those ports, which is the repository's existing rule.

§17 The engine baseline

The twelve invariants of revision 3 stand [R]. The file is the record. Splice-only writes that refuse when a range is ambiguous. Content-addressed versions. Every change carries an author and an intent. A proposal is a first-class object. Everything has a stable address. Machine-readable exits: `Accept: text/markdown`, a `.md` twin, a manifest and

checksums. A capability surface, not a screen surface, so the MCP server, the command line and the API are thin adapters. Permissions attach to the operation. Budgets and breakers sit at the operation layer. Deterministic rendering, and any block we invent degrades to readable text. No silent merge, ever.

Two measured defects are fixed before any public claim [0]: a column-zero list item in front matter refuses 83 percent of real vaults, about four days of work, and a trailing comment is deleted on a set. The audit read the engine and confirmed both, and names a third, unlabelled defect in its A16; section 25 carries it.

Review state is dropped [Z] (F058). The 9 September plan's headline was a per-span read state in a sidecar, `.frontmatter/review.jsonl`, that never existed in the code and failed an adversarial round. The 13 September reset replaced the product, and this plan's S20 is a change queue: who changed what, accept or reject, one by one. Attribution survives as a mark in the version record, section 20. CLAUDE.md's "Review state" paragraph is rewritten on this branch to say so. Almanac, which shipped read receipts and shut down, is the precedent section 25 records.

What this makes the product [P]. The vault is the agent's memory. The map is its index. The instruction file is its policy. The blueprint is its brief. The change queue is how it is supervised. Doc mode, the flow view, the portfolio and Drive sync are all projections and adapters over the same bytes, which is why none of them needed a new format.

§18 The data model

The entities the features imply, where each lives, and its rules [P]. Sizes are per record; retention is what the person can rely on; deletion is what happens on account deletion.

Entity | Lives in | Size and limit | Retention | On account deletion | Who can read Account | Firebase Auth and a profile document | one | until deleted | removed within 30 days | the person Workspace and project | Postgres rows | 50 cloud documents on Free | until deleted | removed | owner and members by role Document head | Postgres row pointing at the current version key | one per document | until deleted; 30 days in trash | removed | by role Version | R2 object keyed by document id and content hash | 5 MB a file on Free, 25 MB on Pro | 7 days on Free, 90 on Pro, then pruned to the head | removed | by role Upload | R2 object under the document | 1 GB an account on Free, 10 GB on Pro | with the document | removed | by role Share link | Postgres row: token, role, expiry, password hash | one per link | until expiry or revocation | removed | anyone with the token Published page | Postgres row and a rendered R2 object | 5 on Free | until unpublished | removed and the URL goes dark | anyone Collaborator | Postgres row: account, document, role | 3 live on Free | until removed | their rows removed, documents stay with the owner | owner Comment | Postgres row keyed by document and content hash | text only | with the document | removed | by role Change queue item | Postgres row: author, source, span, proposed bytes | one per proposal | until accepted or rejected, then a version | removed | by role Idea, decision, blueprint | markdown files in the project, plus a Postgres row for state | fifteen files a blueprint | as documents | removed | by role Template | markdown files, ours or the person's | one folder | ours forever, theirs as documents | theirs removed | the person, or anyone for ours Connection | Postgres row with the encrypted OAuth or installation token | one per provider | until disconnected | revoked at the provider and removed | the person Agent token | Postgres row with a hash and a scope | many | until revoked | revoked | the person Ledger entry | Postgres row: account, kind, delta, model, cost | one per call or grant | 180 days, then aggregated | aggregates kept without the account id | the person Plan and invoice | Postgres rows and Razorpay's records | one per period | as the accountant requires, at least the statutory period | kept as the law requires, unlinked from the profile | the person Security log | an append-only store in Mumbai | one line per event | 180 days rolling | kept for the period | the operators Local draft | IndexedDB or the origin private file system, and the desktop folder | the device's quota | until synced or evicted | not ours | the device

Export. Everything under Documents, Ideas and Settings exports as files a stranger's tool reads: markdown, the uploads, the versions as a folder of files, comments and change-queue items as a JSON file beside each document, decisions and blueprints as they are. Section 25 records that the review sidecar of the earlier plan is not among them, because it does not exist.

§19 Permissions and roles

Role | Read | Edit | Propose | Apply | Comment | Share | Publish | Export | History | Invite | Delete Owner | yes | yes | yes | yes | yes | yes | yes | yes | yes Editor | yes | yes | yes | yes | yes | no | no | yes | yes | no | no Commenter | yes | no | no | no | yes | no | no | no | no | no Viewer | yes | no | no | no | no | no | no | own copy | no | no | no Link, read | yes | no | no | no | no | no | own copy | no | no | no Link, edit | yes | yes | yes | yes | yes | no | no | yes | no | no | no Published-page reader | yes | no | no | no | no | no | no | own copy | no | no | no Agent token | yes | no | yes | never | no | no | never | yes | yes | no | no

A collaborator on Free sharing a Pro owner's document has the owner's limits, not their own. Ownership transfers to another account on request, never by an agent. A revoked collaborator keeps nothing but the exports they already made. Sharing a project shares every document in it at the same role; sharing a document shares that document alone.

§20 Format specifications

Every format the plan invents or adopts, with a version field, a rule for unknown fields, a stated degradation in a plain markdown reader, and a test [P] (A34, F050). The full text of each lives in specs/formats/ from phase A.

Format | Version field | Unknown fields | In a plain reader | Test fm-chart, fm-flow, fm-draw and any fm- block | the info string, fm-chart@1 | ignored, kept | a code block; the table above a chart stays a table | a fixture per block, byte-exact round trip Doc mode front matter keys (title, subtitle, page, margins) | frontmatter: 1 | ignored, kept | front matter, invisible in most renderers | export fixtures Flow view conventions | none, they are headings | n/a | headings and lists | a fixture document MANIFEST.json | "version": 1 | ignored | a JSON file | a schema test SHA256SUMS | the format's own | n/a | a text file | verified on every publish; the root hash printed on the page DECISIONS.md | decisions: 1 in front matter | ignored | a document with one heading per decision and an open marker for Not sure | a fixture MAP.md and graph.json | graph: 1 | ignored | a document and a JSON file | rebuilt from files, compared to the fixture The review sidecar | dropped; see section 17 ||| The AI mark | in the version record: author, source (person, ai, agent), model, ask, accepted_by, at | ignored | nothing in the file by default; if the person turns inline marking on, an HTML comment <!-- ai: model, ask --> beside the span, invisible when rendered, visible in source | a fixture with and without inline marking; export strips the comment unless asked to keep it The portfolio front matter | portfolio: 1 | ignored | front matter; the page still renders as a document | a fixture The kit tarball | MANIFEST.json version | n/a | a folder | the kickoff test in section 28

When a specification changes after kits exist, the version field rises, the old version is still read, and the map shows which version a kit carries. SKILL.md follows the skills guide's size limits and AGENTS.md follows agents.md; both are checked by the health panel.

§21 Performance and reliability targets

Targets of our own, measured in the pilot, with the sources they come from [M] [P] .

Measure | Target | Source of the threshold Keystroke echo | under 100 ms | Nielsen's 0.1 s AI request acknowledged | under 1 s; first token under 3 s | Nielsen's 1 s, Doherty's 400 ms A blueprint written | progress shown past 10 s; done in under three minutes on Low | Nielsen's 10 s Save latency, web | under 500 ms to the server on 4G | our own Sync latency, Drive | within five minutes | section 11 First load on a mid-range Android over 4G | LCP under 2.5 s at p75, INP under 200 ms, CLS under 0.1 | web.dev Editor with a 10 MB document | opens under 3 s, keystroke under 100 ms | our own, measured in the pilot Import of 2,000 files | under two minutes, byte-exact | our own Bundle | a real budget replaces the npm run budget echo: 250 KB of JavaScript on the first load of the editor, measured in CI | our own Availability | 99.5 percent a month; the browser keeps every draft when we are down | our own Durable Object round trip from Mumbai | measured in the pilot; over 250 ms is a finding | section 15

§22 The financial model

Every price read from the provider's page on 17 September; every line shows its arithmetic. SIMULATED: computed from the caps and list prices, not from live usage [0] [M] .

Inputs. Pro gross ₹299, GST 18 percent: $299 / 1.18 = ₹253.39$. Razorpay 2 percent plus GST on the fee, 2.36 percent of ₹299 = ₹7.06. Net ₹246.33, which is \$2.57. Annual ₹2,499 nets ₹171.57 a month. A Pro user at full caps costs \$1.99 in model fees on the default routing (section 14). A free user at full caps costs \$0.0194 on paid Cloudflare neurons once the free pools are exhausted, and nothing before. Fixed a month: one Vercel seat \$20, the domain at \$15 a year and Apple at \$99 a year, together \$29.50; a second Vercel seat adds \$20; the Windows certificate is unpriced. Firestore and R2 are usage rather than subscription, about \$2 and under \$1 at 1,000 users, and sit outside this line as they always did. Revision 5 carried \$54.50 here because Supabase Pro was \$25 of it; the stack decision removes that subscription, and every row below is recomputed.

Monthly result at full-cap use, one seat, default routing.

Signed-in users | Pro at 1 percent | Pro at 2 percent | Pro at 5 percent 1,000 | 10 Pro: \$25.67 revenue, \$19.90 Pro AI, \$19.21 free AI, \$29.50 fixed, result -\$42.94 | 20 Pro: \$51.34 - \$39.80 - \$19.01 - \$29.50 = -\$36.97 | 50 Pro: \$128.35 - \$99.50 - \$18.43 - \$29.50 = -\$19.08 10,000 | 100 Pro: \$256.70 - \$199.00 - \$192.06 - \$29.50 = -\$163.86 | 200 Pro: \$513.40 - \$398.00 - \$190.12 - \$29.50 = -\$104.22 | 500 Pro: \$1,283.50 - \$995.00 - \$184.30 - \$29.50 = +\$74.70 100,000 | 1,000 Pro: \$2,567 - \$1,990 - \$1,920.60 - \$29.50 = -\$1,373.10 | 2,000 Pro: \$5,134 - \$3,980 - \$1,901.20 - \$29.50 = -\$776.70 | 5,000 Pro: \$12,835 - \$9,950 - \$1,843 - \$29.50 = +\$1,012.50

The same at half use (Pro AI \$1.00, free AI \$0.0097), re-derived line by line at the new fixed cost: 1,000 users at 2 percent, \$51.34 - \$20.00 - \$9.51 - \$29.50 = -\$7.67; 10,000 at 2 percent, +\$188.84; 10,000 at 5 percent, +\$661.85; 100,000 at 2 percent, +\$2,153.90.

What the table says [0] . At full-cap use the product loses money below about 5 percent conversion at any scale, and the loss is the free users' model cost, not the fixed costs. At half use it breaks even around 1,350 signed-in users at 2 percent: each such user nets \$0.021834 a month, and $\$29.50 / \$0.021834 = 1,351$. The stack decision moved that point, because revision 5 needed 2,496 users to clear \$54.50. Three levers move it: the routing (Sonnet for everything turns every row negative), the free caps (5 edits instead of 10 cuts the free line by a third), and prompt caching, which is not counted. Break-even Pro users against fixed costs alone: $\$54.50 / \0.58 margin at full use = 94; at half use $\$54.50 / \$1.57 = 35$. The founders' time is not in these lines; at ₹1,500 an hour, 1.2 days a week is about ₹62,000 a month.

A High blueprint adds an assumed research pass of 150,000 tokens in and 40,000 out on Sonnet, \$0.70 (₹67) on top of the blueprint; five High a month is ₹504 of model cost against 15 credits, three top-ups at ₹149 = ₹447 gross. High is priced under its cost at list; the batch API halves it, and the credits price is a founder question if High ships before the pilot measures it.

§23 The legal floor

Holding documents, publishing strangers' pages and taking money bring this in before the first stranger and the first rupee [L] . Each row has an owner and a date. Statutory details are marked unverified where no primary page could be opened from this network on 17 September; the audit's readings are given with their URLs in section 31. Placeholder pages for the four public routes serve on this branch and say the text is pending and due by 15 October 2026.

Due | What | Owner | Date | Verified Before the first stranger | Privacy notice, terms, consent wording; the age floor stated (default eighteen, founder question 12) | Sagnik | 15 Oct 2026 | DPDP Act 2023 section 9 read by the audit on indiacode. Commencement opened by this revision from the Gazette: G.S.R. 843(E) of 13 November 2025, CG-DL-E-14112025-267647, brings sections 7 to 10, section 9 among them, into force "eighteen months from the date of publication of this gazette", published 14 November 2025, so 14 May 2027. The Rules, G.S.R. 846(E), CG-DL-E-14112025-

267650, rule 1: rules 1, 2 and 17 to 21 at once, rule 4 after one year, rules 3, 5 to 16, 22 and 23 after eighteen months. The pilot runs before the duty bites; the terms carry the floor from day one Before the first stranger | Named grievance officer with a published address on every public page; 24-hour acknowledgement, 15-day resolution, 72-hour and 36-hour removal clocks; a report link on S18 | Sagnik as officer of record | 15 Oct 2026 | IT Rules 2021 rule 3 as the audit read the MeitY PDF; not re-opened by this revision Before the first deploy that holds a document | Breach contact filed; a six-hour incident runbook; an append-only security log kept 180 days in Indian jurisdiction | Amit | 15 Oct 2026 | CERT-In directions of 28 April 2022 as the audit read them; not re-opened Before the first rupee | GST position confirmed by a chartered accountant; the price shown GST inclusive; invoice lines | Sagnik | 31 Oct 2026 | Opened by this revision on CBIC's tax information portal, since cbic-gst.gov.in refused curl: Notification 11/2017-Central Tax (Rate) of 28 June 2017, serial 22, "Heading 9984 Telecommunications, broadcasting and information supply services", central tax 9, so 18 percent with the state half; the two 2025 amendments, 05/2025 of 16 January and 15/2025 of 17 September, carry no entry for heading 9984. The single 2026 rate notification and the chartered accountant's reading of which heading a subscription editor falls under remain open Before the first rupee | Razorpay mandates under ₹15,000 and one attempt on Indian cards; refund and cancellation page | Amit | 31 Oct 2026 | RBI/2022-23/73 of 16 June 2022 re-opened by this revision Before the first rupee | Processor agreements: Cloudflare, Google for Firebase Auth and Firestore, Anthropic, Razorpay; each provider's terms for commercial use read | Amit | 31 Oct 2026 | Opened by this revision, not a lawyer's reading. Cloudflare's self-serve agreement, clause 2.2.1(h): you may not "process or collect personal or business credit card information on any web property that is receiving Free Services", so card entry stays on Razorpay's own checkout or the zone moves to a paid plan; its developer platform terms: "Unless otherwise agreed, Cloudflare does not use any Customer Content to train generative AI tools". Supabase's terms restrict reselling the Services themselves, not building a paid product on them. The GitHub Marketplace Developer Agreement "sets forth the terms that govern a Developer publishing Listings on GitHub Marketplace", and the plan lists nothing there. Firebase's terms page did not render outside a browser, and Firebase leaves the stack in phase A Before the first stranger | Accounts moved to the company: the Cloudflare zone, the domain, Razorpay | Sagnik | 15 Oct 2026 | section 24 Before Razorpay goes live | A trademark search for "frontmatter" on the Indian register by a founder (it needs an OTP login); the Front Matter CMS collision at 82,819 installs recorded | Sagnik | 31 Oct 2026 | Unverified; question 7 Before the pilot | EU sign-ups blocked, or a representative engaged | Sagnik | 15 Oct 2026 | carried from revision 3 Before the pilot | A counsel question on the Rights of Persons with Disabilities Act section 46 with rule 15; WCAG 2.2 AA adopted meanwhile, and IS 17802 (Part 1): 2021 tested to (Part 2): 2022 added as the target the draft rule names | Amit | 31 Oct 2026 | Section 46 read by the audit. Rule 15 opened by this revision through the Gazette (F075): the draft Rights of Persons with Disabilities (Amendment) Rules, 2026, CG-DL-E-23072026-274669 of 23 July 2026, replace rule 15(1)(c) with accessibility standards for "websites, mobile applications, tablet applications, other touch-based applications, softwares" made available "to persons in India for public or consumer use, whether such establishment is located within India or outside India", on IS 17802, with an Accessibility Conformance Report "in both human-readable and machine-readable formats". A draft under consultation, not yet law Before the pilot | Consumer Protection (E-Commerce) Rules 2020, rule 4 duties built in whether or not counsel says they bind: a nodal contact resident in India, the legal name and address on the platform, a grievance officer who acknowledges in 48 hours and resolves in a month, consent by explicit action and never a pre-ticked box, refunds within the RBI's period, and the National Consumer Helpline partnership | Sagnik | 31 Oct 2026 | Opened by this revision from the Gazette, since consumeraffairs.nic.in refused every route. G.S.R. 462(E), CG-DL-E-23072020-220661: rule 2(1)(a) covers "all goods and services bought or sold over digital or electronic network including digital products" and rule 3(1)(b) defines an e-commerce entity as "any person who owns, operates or manages digital or electronic facility or platform for electronic commerce". The amendment of 10 September 2026, CG-DL-E-10092026-276125, turns the helpline partnership from an endeavour into a duty. Whether a subscription editor is an e-commerce entity is a counsel question Now | Third-party notices: the OFL texts and the Apache notices in THIRD-PARTY-NOTICES.md; the OFL line atop the screens' font file; Mosvita no longer embedded in the PDFs | done on this branch | 17 Sep 2026 | OFL texts for Google Sans Code and Google Sans Flex opened; the "Google Sans" family's own licence file unverified (F070, F073)

§24 Ownership and accounts

Asset | Held by today | Moves to | By GitHub repository `studiozephyrus/frontmatter` | the studio's user account | the company's organisation | before the pilot Vercel project | team `zsco` | the company's team | before the pilot Firebase project `frontmatter-md` | a studio Gmail account | retired in phase A | phase A Cloudflare zone `frontmatter.in` | a founder's personal account, by deliberate choice | the company | before the first stranger Domain registration | unverified | the company | before the first stranger Razorpay | not opened | the company | before the first rupee Apple Developer Program | not opened | the company | phase F Model provider accounts | not opened | the company | phase B Analytics and error accounts | unverified | the company | phase A Intellectual property between the founders | unverified | a written agreement | before the first rupee

§25 Closure of the earlier findings

Finding | Where recorded | State in this plan Almanac shipped read receipts, raised \$45 million, shut down 31 January 2025 | 9 September briefing | Named here for the first time; the review-state product is dropped, section 17 Zed Delta stakes span attribution on CRDTs; a written rebuttal is owed | CLAUDE.md, card E35 | Still owed; due before phase D, section 15 The attention literature is hostile to an unreviewed-percentage badge | 9 September briefing | No such badge in this plan A research agent fabricated VS Code quotations | 9 September briefing | Every quotation re-fetched by the audit; two shortened ones restored in full (F042) Review state does not exist in code; the engine is unwired | 9 September briefing | Review state dropped; the engine's wiring is phase B; the two named defects and the audit's third are fixed before any public claim `GITHUB_REPO` defaults to a sibling project's vault | 9 September briefing | Fixed in Phase 0, as revision 3 had it The twenty hand-made kits gate | 13 September plan | Restored as Phase 0 (F009) The extension form factor | earlier plans | Not planned Bring-your-own key | card F10 | Founder question 10 (F056) The community | revision 3 | Later, named (F023) The Max tier | revision 3 | Later, named (F023) The 8 September gap register's Firestore contradiction | GAPS-2026-09-08 | Settled by the founders on 17 September: Firestore is the database, and firestore.rules is live work, so the removal proposed as F045 is withdrawn The shipped app's local drafts under legacy keys | AGENTS.md section 8 | Migrated in phase A, section 15 (A33) The image proxy and the delimited data block | revision 3 security | Restored, section 14

§26 The build, in appetites

The dev plan follows approval. Its shape, in fixed-time phases with variable scope `[M]` Shape Up. Question 1 decides the pace and therefore which phases are Later.

Phase | Appetite | What ships 0. Before code | 2 weeks | The legal floor's first rows; the accounts moved; the public pages written; the pace published every Friday; the `GITHUB_REPO` default fixed; the format specifications drafted; twenty blueprints made by hand for twenty people outside the studio, watched for whether five run the kickoff and two of ten edit a kit again A. The door and the home | 3 weeks | Sign-in through Firebase Auth and Auth.js as already shipped, firestore.rules hardened from prototype to product, the local drafts migrated; Home, settings, plan page; the entitlements layer, the usage ledger and the configuration panel of section 30; Firestore and R2 adapters B. The editor as shipped, plus Doc mode | 4 weeks | The workspace on the new stack; the two engine defects and the audit's third fixed with red proofs; Doc mode with the 20 and the 15; problems and formatter; the AI box and menu on the free chain with the breaker; the `--ai` token and the code face in `globals.css`; bring-your-own key if question 10 says so C. Ideas | 4 weeks | The ideas tab, Low, the fifteen-file blueprint, the consistency check, the unlisted link, the kickoff prompt with the out-of-band hash, the map; gated on Phase 0's result D. Sharing | 3 weeks | People with the matrix, links with expiry, published pages with the `.md` twin, the footer and the grievance route; the change queue; history; live editing on Durable Objects under section 15's rule, or Later if question 8 says so; the Zed rebuttal written E. In and out | 3 weeks | Folder upload, Obsidian and Notion import, Google Docs and Word with the 10 MB refusal, the GitHub

App, Google Drive sync at a five-minute poll F. Everywhere | 3 weeks | Offline in the browser, the desktop app on the new stack built on per-platform CI runners, signed for macOS at 99 USD a year, the phone layouts, quick capture, dark mode G. Views and blocks | 3 weeks | Flow, slides, mind map, Excalidraw, Mermaid types, KaTeX, templates, tasks, calendar H. Pro | 2 weeks | Razorpay with the mandate rules, Medium and High, the Claude routing of section 14, password links, 90-day history Later | | Kanban and table-to-chart blocks, the portfolio, the MCP server and API with the agents card, Team, Max, the community, a custom domain, Notion API import, a signed Windows build

Twenty-seven weeks of appetite at full time, one more than revision 5: phase A grew by a week for the configuration panel, and shrank by the Supabase migration it no longer has to do. At the measured pace, which the audit recomputed at 0.93 to 1.21 days a week, that is 99 to 129 calendar weeks. The default in question 1: phases O, A, B, D and H at the measured pace with dates published every Friday; E, F and G Later; a contractor for D and F if the pace has not doubled by the pilot. Content is costed too: the seven templates with their question banks, the consistency checks, the kickoff prompts, help text and empty states are about thirty days of writing the earlier plan did not count (A44).

§27 Risks

Risk | What we do Sign-in first costs sign-ups | Measure the drop between the sign-in page and the first save. If it is over a third, publish an editor-first path for shared pages only Obsidian ships Multiplayer, listed as Planned on its roadmap | Our live editing competes with an announced feature; sharing by link, the change queue and the blueprint carry the collaboration story, and question 8 may cut live editing until a pilot asks Obsidian for Work, listed as Active | Their team story lands before ours; Team stays after Pro and the plan says so Obsidian ships a web version | Their third most-liked request, not on their roadmap. Our opening narrows to AI, collaboration and the blueprint Obsidian takes the agent position | Its chief executive's skills repository has 48,440 stars. We win on the editor a person pays to open, on collaboration, and on the brief A free provider changes its terms | The chain has four providers and the desktop has a local model. Terms are re-read monthly and the date is recorded, with the RBI circular Free-tier abuse of the model pools | Pools are per organisation, so one abuser drains everyone. Per-account budgets, a breaker, sign-in first, and bring-your-own key if question 10 says so Pro loses money at full use | The default routing keeps ₹55 a user; Sonnet for everything loses ₹137; the price and routing are question 2 and caching is measured in week one Safari evicts local drafts | The banner says so, persist is requested, the first connection pushes, the home-screen install exempts the app The Google OAuth app or the GitHub App is suspended | Every user is locked out or the write-back dies; an email magic link is the fallback sign-in (question 10 and section 15) and exports never need a connection The name collides with Front Matter CMS | The search is run before Razorpay goes live; the screens change in a day if the name changes Windows cannot be signed from India | Windows is shown as coming; a commercial certificate is priced before phase F The scope is too large for the pace | The Later column is the release valve. Nothing in phases O to B is optional

§28 What we measure, and the pilot

Definitions [P] (F067). An active user opened a document they own on two distinct days in seven. A finished blueprint reached Hand off with all fifteen files present and the consistency check passing. A conversion is a Razorpay mandate approved, not a click on Pro. An accepted proposal is an item accepted individually in the change queue; Accept all counts separately. An imported file counts as byte-exact only when the bytes compare equal after a round trip.

Measures. Signed in to first save under two minutes. Documents per active user. Blueprints started and finished, by depth, and edited again within seven days. Imported vaults, and the share of files that imported byte for byte. Published pages and the sign-ups they bring. Free to Pro conversion, and which cap tripped first. Proposals accepted against rejected, individually against Accept all. Model spend per active user against the pools. Sync conflicts shown against merges attempted, which must be zero.

The pilot. Twenty people: ten Obsidian users with a vault of at least 200 files who have posted in the web-version or sync threads, five founders or product people who have run Claude Code, Cursor or Codex on a project in the last month, five people who write in Google Docs and share by link. Recruited by a founder's own posts and the studio's network; each receives a hand-made kit for a real idea of theirs, which is the offer that earns the hour. The first five minutes are scripted: sign in, import a vault or write a document, share it by link, run one AI edit and accept or reject it; group one fetches the kit and runs the kickoff. The first week is open with three prompts on days 2, 4 and 7.

Stop. Fewer than four of twenty active in week two. Fewer than two of ten kit recipients run the kickoff. Fewer than three of twenty name the problem the product solves without being prompted. Any one of the three stops the next phase.

Continue. Six of twenty active in week two. Three of ten run the kickoff and edit the kit again. One person asks how to pay before being told the price. All three justify phase C and the Pro build; two of three justify phase C alone.

§29 Questions for the founders

Decided on 17 September, and written into this revision [Z]: the stack is Next.js with R2 and Firestore, section 15; tier contents are set from the configuration panel, section 30. **Deferred on purpose:** the tagline, the positioning and the product-market read, to be settled while phase A is built rather than before it.

What the configuration panel absorbs. Eleven of the eighteen below stop being decisions the build waits on, because the panel sets them at run time: the Pro price and routing, the provider chain, the free caps, the age floor's number, the pilot thresholds, the grievance officer's name, the indexing default, and the four feature flags. They still need answers before the pilot meets a stranger. They no longer need answers before phase A starts.

What the panel cannot absorb, and still gates the build: which bytes we hold and from which phase, question 6, because it is an architecture; the one-sentence definition, question 5, because it decides build order; the pace, question 1; the twenty-kit gate, question 14; the desktop's timing, question 11; the name, question 7; and the accounts that move to the company, question 13.

The sixteen in docs/mvp0/DECISIONS-FOR-FOUNDERS-2026-09-17.md, each with a default this revision is written to. In order: the pace and which phases are Later; the Pro price and routing; the free chain and the sign-in promise; the legal floor and the grievance officer; K1, what the product is in one sentence; K2, which bytes we hold and from which phase; K3, the name; live editing and the CRDT ban; the free caps and the downgrade; bring-your-own key; desktop before or after sync and who signs Windows; the age floor; the accounts that move to the company; the twenty hand-made kits; the pilot's stop and continue lines; which earlier positions stand. And one more, from the audit's growth angle: whether published pages should be indexed, since not indexing removes search as a channel and the plan has no other.

Where this revision narrowed a founder ask, and says so [Z]. "Every feature free, quantities capped" became "every editing feature free; password links, Medium and High, the portfolio and branding removal are Pro" (F003). The free caps are the market's 50, 5 and 3, not the candidates' 5, 2 and 1. K1, K2 and K3 were taken implicitly by revision 4 and are now defaults with the cards' recommendations beside them: K1 rec b, K2 rec b, K3 rec c in decisions/v2.

§30 The configuration panel

Why it exists [Z]. The founders decided on 17 September that what a tier allows is set from a panel, not from constants in the source. That turns most of the open questions from decisions before the build into settings after it: eleven of the eighteen on the decision sheet become rows in a table this panel edits. It is built in phase A, before the features that read it, because retrofitting it later means finding every hard-coded cap in the product.

What it sets [P]. Every row is read at run time. Nothing here needs a deploy.

Setting | What it holds | Read by Plan limits | Documents, published pages, live collaborators, history days, upload size and total, AI edits, blueprints, repositories, pushes, per plan | Every cap check, through one function Prices | The monthly and annual price, per plan, and the top-up prices | The plan page and the Razorpay call Model routing | Which model serves edits, documents and blueprints, per plan | The AI router of section 14 Provider chain | The free chain's order, and whether each provider is enabled | The free-tier router Feature flags | Live editing, bring-your-own key, the email magic link, the indexing default | The feature gate Pilot thresholds | The stop and continue lines of section 28 | The measurement dashboard Exceptions | A temporary limit granted to one account, with an expiry | The same cap check

One read path [P]. A single function, `limitsFor(account)`, resolves a plan row plus any exception into the limit set. Nothing else in the product reads a cap. A number that appears anywhere else is a defect, and the architecture gate of `npm run arch` is the place to catch it, the same way it already catches a `process.env` read outside config and infrastructure. The usage ledger of section 18 is the counter each check runs against.

A limit that falls is a downgrade [L]. Raising a limit takes effect on the next read and nobody notices. Lowering one below what an account already holds puts that account into the over-cap state of S33: everything stays readable and exportable, nothing new is created until it is under the cap. The panel says how many accounts a change puts over the line, and names them, before it saves.

What the panel cannot do [L]. Three things stay outside it, because a setting cannot undo a promise:

- **The training promise.** The sign-in page says we never train on documents. That is a claim about which providers are in the chain, not a number, and adding a provider whose terms permit training changes the sentence, not a row. Question 3 on the decision sheet.
- **The age floor, once someone has signed up under it.** The panel can hold the number. It cannot re-consent the people who accepted the old terms. Question 12.
- **Whether bytes are held at all.** That is section 18's architecture and phase A's shape, not a setting. Question 6.

Who holds it [Z]. Super admin is a flag on an account, not a plan, and both founders hold it. It is checked server-side on every write, never in the browser. Every change writes an audit row: who, which setting, from what, to what, when, and how many accounts it moved. That log is read-only in the panel and kept as long as the security log of section 23.

Screens S35 to S38.

§31 Sources

Every page below was opened on the date shown and quoted verbatim in this plan. Pages this revision opened on 17 September to confirm a finding are marked so.

Re-opened by this revision, 17 September. [Craft pricing](#) · [Docmost pricing](#) · [Obsidian roadmap](#) · [GitHub App permissions](#) · [GitHub installation tokens](#) · [RBI circular RBI/2022-23/73](#) · [Google Sans Code OFL](#) · [Google Sans Flex OFL](#) · [OpenRouter privacy and logging](#) · [OpenRouter endpoints API](#) · [DiceBear HTTP API](#) · [Cloudflare website terms](#) · [Supabase terms](#) · [Firebase terms](#) · [frontmatter.in/privacy](#) and the three sibling paths, 307 to /login on the deployed app

Opened by the audit, 17 September, not re-opened here. [IT Rules 2021, MeitY](#) · [CERT-In directions, 28 April 2022](#) · [Rights of Persons with Disabilities Act 2016](#) · [Azure Artifact Signing quickstart](#) · [Apple Developer Program](#) · [Tauri Windows installer](#) · [Durable Objects data location](#) · [R2 data location](#) · [Semantic Scholar licence](#) · [MCP governance](#) · [Notesnook pricing](#)

Free tiers and metering, 17 September. [Google Docs](#) · [Google storage](#) · [Notion pricing](#) · [Superhuman Docs](#) · [Bear](#) · [Obsidian](#) · [HackMD](#) · [Dropbox Paper](#) · [Slite](#) · [Nuclino](#) · [Outline](#) · [GitBook](#) · [Confluence](#) · [Anytype](#) · [AFFiNE](#) · [Notesnook](#) · [Joplin Cloud](#) · [Simplenote](#) · [Standard Notes](#) · [Capacities](#) · [Mem](#) · [Tana](#) · [Reflect](#) · [UpNote](#) · [Evernote](#) · [Tolaria](#) · [CodeGuide](#) · [ChatPRD](#) · [Kiro](#) · [Cursor](#) · [Claude](#) · [ChatGPT](#) · [Canva](#) · [Figma](#) · [Linear](#) · [Vercel](#) · [Loom](#)

The stack, 17 September. [R2 pricing](#) · [D1 pricing](#) · [Durable Objects pricing](#) · [Workers pricing](#) · [Firestore pricing](#) · [Firestore quotas](#) · [Firebase pricing](#) · [Supabase pricing](#) · [Supabase regions](#) · [Neon](#) · [Turso](#) · [PlanetScale](#) · [Vercel Hobby](#) · [Vercel Blob](#) · [Clerk](#) · [Auth.js](#) · [Liveblocks](#) · [PartyKit](#) · [Algolia](#) · [Meilisearch](#) · [Typesense](#) · [Resend](#) · [Sentry](#) · [PostHog](#)

Model providers, 17 September. [Groq rate limits](#) · [Groq models](#) · [Groq data](#) · [Groq services agreement](#) · [Gemini rate limits](#) · [Gemini pricing](#) · [Gemini terms](#) · [OpenRouter limits](#) · [OpenRouter models](#) · [OpenRouter privacy](#) · [GitHub Models](#) · [NVIDIA build](#) · [NVIDIA terms](#) · [Cerebras rate limits](#) · [Cerebras terms](#) · [Together pricing](#) · [Together privacy](#) · [Mistral pricing](#) · [Mistral usage limits](#) · [Mistral terms](#) · [Hugging Face providers](#) · [Workers AI pricing](#) · [Workers AI limits](#) · [Workers AI data](#) · [SambaNova rate limits](#) · [SambaNova pricing](#) · [Anthropic pricing](#) · [Anthropic rate limits](#) · [Ollama](#) · [Ollama FAQ](#)

Doc mode, 17 September. [Google Docs help](#) · [Google Docs formatting not supported in Markdown](#) · [CommonMark 0.31.2](#) · [GFM](#) · [Pandoc manual](#) · [Tiptap extensions](#) · [Tiptap markdown](#) · [Typora support](#) · [Bear Lettera](#) · [Milkdown](#) · [tui.editor](#) · [Obsidian editing modes](#)

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Carried from revision 3, opened 15 and 16 September. [Obsidian feature requests](#) · [Obsidian plugin stats](#) · [Obsidian sync help](#) · [kepano/obsidian-skills](#) · [Claudian](#) · [Tolaria](#) · [AGENTS.md](#) · [MCP specification](#) · [Skills guide](#) · [llms.txt](#) · [mem0 pricing](#) · [Zep pricing](#) · [Anthropic API pricing](#) · [Razorpay pricing](#) · [Razorpay subscriptions and mandates](#) · [mammoth](#) · [Drive export formats](#) · [decisions/questions.js](#), 210 cards